Judged by the Company You Acquire

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Abstract

An acquirer's decision to undertake an acquisition, together with its choice of target, can systematically reveal its technological gap. The underlying selection mechanism allows the market to distinguish technological laggards from leaders and update valuations accordingly. I measure this revealed gap as the technological overlap between the target and the acquirer's industry. Acquirers with larger revealed gaps experience more negative announcement returns, but only when the target is more transparent than the acquirer and thus conveys information beyond the market's prior beliefs. This effect is concentrated in the component of target choice unexplained by acquisition determinants identified in prior studies. Moreover, revealed gaps induce opposite market reactions among the acquirer's industry peers. Additional tests help rule out alternative explanations. Taken together, these findings highlight a selection-based mechanism that helps explain the cross-section of acquirer announcement returns.

Keywords: Mergers and acquisitions, revaluation, technological gap, target selection

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1 Introduction

A long-standing puzzle in the mergers and acquisitions (M&A) literature is that acquirer announcement returns are, on average, neutral to negative, yet acquirers do not systematically underperform post-acquisition.[1] This challenges the widely adopted interpretation that announcement returns primarily reflect deal quality. Adding to this puzzle is a stark cross-sectional difference in announcement returns between acquirers of public and private targets: negative for the former, but positive for the latter.[2] This paper documents an important component in acquirer returns that is consistent with these patterns and arises from the interaction of selection and information.

Acquisitions are endogenous to firms' resource constraints, including technological needs. I theorize that technological laggards self-select into acquisitions to close technological gaps relative to their industry peers, as acquisitions become more cost-efficient as technological needs grow.[3] In turn, the decision to acquire, together with the choice of target, reveals the acquirer's weak technological standing. Although such acquisitions maximize the acquirer's intrinsic value, the revelation they trigger corrects the market's prior overvaluation of the acquirer, thereby reducing acquirer announcement returns on net.

IBM's 2018 acquisition of Red Hat illustrates this tension. Some observers emphasized the strategic fit and synergies of combining IBM with Red Hat's cloud technology, whereas others interpreted the deal as an admission that IBM had failed to develop comparable cloud capabilities internally.[4] This paper argues that both perspectives can be valid at the same time. When the market has incomplete information about the acquirer's technological standing, the technology that makes a target attractive may also reveal that the acquirer is seeking to close a technological gap. As a result, acquisitions that increase the acquirer's intrinsic value may still trigger negative announcement returns if they induce a downward revaluation of the acquirer's standalone value.

[1] For empirical evidence on acquirer returns, see Akbulut and Matsusaka (2010), Betton, Eckbo, and Thorburn (2008), and Moeller, Schlingemann, and Stulz (2005). For empirical evidence on acquirers' post-acquisition performance, see Fama (1998), Franks, Harris, and Titman (1991), Mitchell and Stafford (2000), and Savor and Lu (2009).

[2] See Ang and Cheng (2006), Capron and Shen (2007), Chang (1998), Faccio, McConnell, and Stolin (2006), Fuller, Netter, and Stegemoller (2002), and Hansen and Lott (1996).

[3] Jovanovic and Rousseau (2002) propose a similar self-selection mechanism in the context of physical assets, arguing that acquisitions become more cost-efficient as the investment scale increases. They provide suggestive evidence in their Figure 5.

[4] On the positive side, Barclays analysts wrote that "This deal represents the culmination of IBM's existing partnership with Red Hat, and, in our view, allows IBM to gain a highly strategic asset to advance its hybrid cloud initiatives" (quoted in Reuters, 2018, "IBM to acquire software company Red Hat for $34 billion"). By contrast, Ben Thompson wrote that "one of the most important takeaways from the Red Hat acquisition is the admission that IBM's public cloud efforts are effectively dead" (Thompson, 2018, "IBM's Old Playbook," *Stratechery*).

Moreover, this negative effect is more pronounced when the target provides information beyond what was previously available about the acquirer. Accordingly, acquirers of public targets are more likely to experience negative revaluations than those acquiring private targets, as public targets are more transparent and offer richer disclosures. This greater transparency enables more informative revelations and leads to lower acquirer announcement returns in deals involving public targets. Importantly, this mechanism also predicts transparency-driven cross-sectional variation in acquirer announcement returns within the group of public or private targets.

I provide empirical evidence consistent with this mechanism. I measure the acquirer's revealed technological gap as the technological overlap between the target and the acquirer's industry peers. Greater overlap implies that the acquirer's technological needs lie more squarely within the industry's existing technology set and are thus more consistent with a gap-closing motive. To operationalize this measure, I leverage patent data from the U.S. Patent and Trademark Office (USPTO) to characterize firms' technological portfolios based on their patenting activity across technological fields. I then compute the cosine similarity between the technological portfolios of the target and the acquirer's industry peers as my measure of the revealed technological gap.

I draw a sample of 1,044 public-target and 962 private-target acquisitions involving publicly listed acquirers between 1990 and 2020 from SDC. Among public-target deals, a strong negative association between acquirer announcement returns (measured by cumulative abnormal returns over the three- or seven-day window surrounding the announcement) and the revealed-gap measure emerges: a one-standard deviation increase in the gap is associated with a 0.83 to 1.83 percentage-point (11 to 22% standard deviation) decline in acquirer announcement returns—an effect larger in magnitude than that of the method of payment. These findings are consistent with the interpretation that a greater revealed technological gap induces more negative revaluations of the acquirer.

I further hypothesize that the presence of prior information about the revealed technological gap weakens the negative effect. Evidence supports this hypothesis: the negative effect is weaker when the acquirer already holds a toehold stake in the target prior to the announcement, when the deal is rumored, and when it is preceded by a back-to-back acquisition involving the same acquirer.

In contrast to public targets, private targets are more opaque because they face minimal disclosure requirements. Consequently, the market may be less likely to rely on them as a credible source of information when inferring the acquirer's technological standing, leading to a weaker relationship between acquirer announcement returns and the revealed-gap measure. Regression results support this argument, suggesting that differences in transparency may

be a driver of differences in acquirer announcement returns.

Importantly, this mechanism also predicts cross-sectional variation in acquirer announcement returns *within* deals involving public or private targets. I first focus on public-target deals and proxy for firm opacity using pre-announcement bid-ask spreads and analyst following. Using these two proxies, I measure the extent to which an acquirer is more opaque than the target. A larger opacity difference might lead the market to more heavily rely on the target, relative to the acquirer, as a source of information to infer the acquirer's technological standing. The evidence shows that the negative effect occurs almost exclusively when the acquirer is more opaque than the target, and it becomes stronger as the opacity difference widens.

I then shift the focus to private-target deals. I collect data on their prior bond issuances and late-stage VC funding rounds before the acquisition announcement. I argue that such private targets are substantially more transparent than other private firms because they face heightened disclosure demands from creditors or VC investors. Within this select group of private targets, I find evidence of negative revaluations. These findings strongly support the hypothesized role of informational frictions in shaping the cross-section of acquirer announcement returns.

Does the revealed-gap measure really capture unexpected information beyond the market's prior beliefs? To answer this question, I decompose the revealed-gap measure into two components: one predicted by acquisition determinants documented in the literature, and a residual orthogonal to them. The residual isolates the part of target choice not predicted by the included observables. Regression results show that the negative effect is entirely attributable to the residual, consistent with new information revelation.

Lastly, I turn to the acquirer's industry peers. When acquisition announcements reveal technological laggards, they simultaneously reveal possible leaders. Accordingly, industry peers that do not pursue acquisitions may experience positive revaluations. The regression results support this hypothesis. I further rule out the alternative explanation that the revealed gap captures industry-wide shocks that alter the probability that firms will become future acquirers or targets.

As an extension, I show that the effect persists for at least one year post-announcement with no evidence of a reversal. For robustness, I rule out several alternative explanations, including dissynergies arising from acquirer-target mismatch, overpayment, and partially priced-in acquisition gains due to pre-announcement anticipation. I also demonstrate that the main results are not sensitive to alternative definitions of the revealed-gap measure that vary the granularity of the patent classification scheme, the length of patenting history, or the weighting of patents by quality. In a further extension, I explore the reverse scenario: whether

the market infers the target's technological gap from the acquirer's observable technological profile, and find only weak and suggestive evidence.

Taken together, these findings help explain the cross-section of acquirer returns. The evidence is consistent with a neoclassical framework in which acquisition announcements reveal unobserved technological standing under informational frictions, with the relative opacity of the acquirer and the target shaping the strength of this revelation.

The rest of the paper is organized as follows. Section 2 presents a formal model and derives testable predictions. Section 3 describes the data, outlines the empirical strategy, and defines the key variables. Section 4 reports the main empirical results and discusses the findings. Section 5 addresses alternative explanations, conducts robustness checks, and explores extensions. Section 6 concludes.

Related literature. This paper is related and contributes to several strands of the literature. First, it contributes to the literature on revaluation in M&As. The most closely related study is Wang (2018), which models M&A transactions as a process of reallocating complementary assets in a two-sided search market and structurally decomposes announcement returns into components reflecting revaluations and synergies. While that framework offers valuable insights, it relies on simplifying assumptions, such as the notion that firms produce with a pair of complementary assets. This paper complements that approach by unpacking the "black box" of revaluations and grounding the analysis in a concrete and economically meaningful context: technological gaps, thereby highlighting technological complementarity as a crucial driver of revaluations in M&As. Besides, Blouin, Fich, and Tran (2020) show that firms acquiring R&D-intensive targets in response to tax incentives may inadvertently signal a lack of internal R&D capabilities. Jacobsen (2014) finds that withdrawing from overpriced M&A deals positively signals CEO quality, prompting positive revaluations. This paper complements these works by documenting a more direct revelation mechanism that emerges in a broader and more commonly observed strategic context.

Hietala, Kaplan, and Robinson (2002) are among the first to decompose acquirer returns into a synergy component and a revaluation component. Earlier studies of acquirer revaluation have primarily focused on the method of payment as the key revelation channel, where stock offers may signal acquirer overvaluation and trigger negative market reactions (Fuller, Netter, & Stegemoller, 2002; Moeller, Schlingemann, & Stulz, 2004; Myers & Majluf, 1984; Shleifer & Vishny, 2003; Travlos, 1987). This paper contributes to that literature by introducing a novel revelation mechanism rooted in the acquirer's choice of target. More broadly, the paper also relates to a growing literature on the revaluation of targets (Malmendier, Opp, & Saidi, 2016) and industry peers (Cai et al., 2024; Derrien et al., 2023) in response to M&A

announcements.

Second, I contribute to research addressing the disparity in announcement returns between public- and private-target acquisitions. For example, Chang (1998) and Fuller, Netter, and Stegemoller (2002) suggest that private sellers receiving stock may become blockholders in the acquiring firm, thereby enhancing post-deal monitoring and increasing acquirer valuation. Ang and Kohers (2001) and Chang (1998) further posit that private targets may accept lower acquisition premiums due to their illiquidity, enabling acquirers to capture more value. Hansen and Lott (1996) argue that in public-target deals, acquirer shareholders can hedge the risk of overpayment by also holding shares in the target—an option unavailable in private-target transactions. Studying a sample of Western European acquisitions, Faccio, McConnell, and Stolin (2006) conclude that none of these theories fully explains the observed acquirer return disparity between public- and private target deals.

Complementary to these existing explanations, my results suggest that the divergence in returns may also stem from the differential informativeness of the target. Public targets, being more transparent, allow the market to extract more inferences about the acquirer than their private counterparts.

Third, this paper relates to a broader literature on acquirer-target complementarity, which explores synergies across dimensions such as product markets (Hoberg & Phillips, 2010; Jia & Sun, 2022), human capital (Lee, Mauer, & Xu, 2018), and assets in a general sense (Rhodes-Kropf & Robinson, 2008; Rhodes-Kropf, Robinson, & Viswanathan, 2005). In a closely related but distinct context, Bena and Li (2014) show that technological overlap between acquirers and targets can generate synergies. This paper emphasizes technological complementarity as a motive for acquisition but highlights an informational implication: target choice can reveal hidden traits of the acquirer that shape the acquisition decision. This perspective opens avenues for future research on the informational dynamics of M&As, particularly in relation to other forms of acquirer-target complementarity.

Lastly, this paper reinforces a growing view that challenges the uncritical use of announcement CARs as proxies for value creation in M&A research (Ben-David et al., 2025). The findings suggest that CARs may not simply be a noisy measure of deal net present value (NPV), but could be systematically biased. In particular, when firms pursue acquisitions to address competitive disadvantages, the market reactions may reflect a downward revaluation of the acquirer's standalone value. As a result, CARs may systematically understate the deals' true NPV.

2 Theoretical framework

In this section, I construct a simple model to formalize the hypotheses of the paper. I first model firms' endogenous decision to close technological gaps through acquisitions. Next, I introduce informational frictions into the framework and analyze how they shape the market's reactions toward acquisition announcements. Finally, I discuss testable predictions of the model and formulate hypotheses.

2.1 Acquisitions as endogenous decisions

To set the stage, consider an economy populated by a large number of profit-maximizing firms. Each firm i is endowed with $z_i \in [0, z^{max}]$ units of technology as a production input, where z^{max} represents the entire technology set of the industry. The cross-sectional variation in this initial technology endowment may arise from technological shocks that reconfigure an industry's technology set, leaving some firms better positioned than others.

Firms produce for a single period and then distribute all proceeds to investors. For simplicity, I abstract away from other inputs. The production function is given by

$$Q(z_i) = \kappa z_i^{\alpha}, \tag{1}$$

where $\kappa > 0$ captures industry-specific productivity, and $\alpha \in (0, 1)$ ensures decreasing returns to scale.

Before producing, firm i can *build* x_i additional units of technology internally for $p > 0$ per unit, or *buy* y_i units externally for $p + f$ per unit through acquisition, where $f > 0$ captures costs associated with the transaction (e.g., search costs and advisory fees).

Securing these technology units takes time. The model does not explicitly model the passage of time, but instead incorporates it through opportunity costs that capture the foregone economic value that could otherwise have been created with the time. The per-unit time cost is fixed at β for externally acquired units, but is convex for internally developed units, taking the form βx_i^{ϕ}, where $\phi > 1$. This specification reflects the concept that R&D projects become disproportionately more complex as their scale increases.[5]

Let $I_i = x_i + y_i$ be the gross investment into technology. The total cost of technological investment is the sum of purchase costs and time costs,

[5]This concept dates back to Brooks's law, which describes the phenomenon whereby adding personnel to a delayed complex project can further delay it because of mounting communication overhead (Brooks, 1975). Huberman and Wilkinson (2005) theoretically demonstrate that the completion time of complex cooperative projects increases with system size because of task interdependencies. Similarly, Boehm (1981) proposes the Constructive Cost Model (COCOMO), in which software development costs are modeled as an exponential function of project size, similar to the specification used in this paper.

$$c(I_i) = \underbrace{px_i + (p+f)y_i}_{\text{purchase costs}} + \underbrace{\beta(x_i^{\phi} + y_i)}_{\text{time costs}}. \tag{2}$$

Given gross investment I, a firm's intrinsic value is thus given by

$$V_i(z_i, I_i) = Q(z_i + I_i) - c(I_i). \tag{3}$$

For simplicity, I suppress the subscript i in the rest of the chapter.

While the model focuses on acquisitions aimed at closing technological gaps, firms also undertake acquisitions for other purposes. These transactions are treated as exogenous and thus fall outside the decision-making process analyzed here. Their implications are deferred to Section 2.2.

2.1.1 Build or buy: a pecking order strategy

The optimal investment level equates the marginal product (MP) of technology with its marginal cost (MC). The MP of technology, given gross investment I, is

$$Q'(z + I) = \alpha\kappa(z + I)^{\alpha-1}. \tag{4}$$

The MC of technology depends on whether the firm builds internally or buys externally. The respective MC of the two options is given by

$$c_x = p + \beta\phi x^{\phi-1}, \tag{5}$$

$$c_y = p + f + \beta. \tag{6}$$

Figure 1(a) plots the two MP functions. The MC of buying (c_y, blue line) is constant, while that of building (c_x, red line) increases with x due to convex time costs. They intersect at the investment level I^*:

$$I^* = \left(\frac{f+\beta}{\beta\phi}\right)^{\frac{1}{\phi-1}}. \tag{7}$$

Since building is cheaper than buying only for investments smaller than I^*, a *pecking-order* strategy emerges: for investments smaller than I^*, the firm builds. Whereas for investments larger than I^*, the firm builds I^* units internally and buys the remaining externally. This

strategy is formalized as:

$$x(I) = \min\{I, I^*\} \tag{8}$$

$$y(I) = \max\{I - I^*, 0\}. \tag{9}$$

The resulting MP of technology is $\min\{c_x(I), c_y(I)\}$, as depicted by the two solid line segments in Figure 1(a).

2.1.2 Firm value under the optimal investment strategy

The optimal gross investment I equates the MC and the MP of technology, which is determined at the intersection of the two curves. Figure 1(b) illustrates two possible scenarios. First, if the curves intersect on the upward-sloping segment of the MP curve ($I \leq I^*$), the firm builds all technology units internally. Second, if they intersect on the flat segment ($I > I^*$), the firm builds I^* units internally and buys the remainder externally. The second scenario gives rise to *gap bidders*. For ease of communication, I label firms in the first scenario as *non-gap bidders*.

The firm's initial technology endowment z determines which scenario applies. Due to decreasing returns to scale, a greater z corresponds to a lower MP of technology (a lower dotted curve). This leads to a smaller optimal investment ($I < I^*$) which is fully satisfied through building. Such firms (non-gap bidders) have technology endowments in the range $z \in [z^*, z^{max}]$ and optimal investment $I \in [0, I^*]$. Conversely, a smaller z implies a higher MP of technology (a higher dotted curve), resulting in a larger optimal investment ($I > I^*$), which requires buying in addition to building. These firms (gap bidders) have technology endowments in the range $z \in [0, z^*)$ and optimal investment $I \in (I^*, I^{max}]$, where I^{max} is reached when a firm has zero endowment. Appendix A.1.1 derives and provides formal expressions for z^*, z^{max}, and I^{max}.

Figure 2(a) plots the final technology level under the optimal investment strategy as a function of z, decomposed into the endowment itself (z, grey), the built units (x, red), and the bought units (y, blue). All gap bidders ($z \in [0, z^*)$) build the same number of technology units ($x = I^*$), while non-gap bidders ($z \in [z^*, z^{max}]$) build fewer as z increases. Besides, gap bidders buy fewer units as z rises.

Notably, all gap bidders converge to the same final technology level:

$$x(z) + y(z) + z = I^* + y(z) + z = I^{max} \quad \forall z \in [0, z^*). \tag{10}$$

This is because they all equate the MP of technology to the MC of buying, which is constant

at $c_y = p + f + \beta$. A key implication is that there exists a bijection between a gap bidder's technology endowment z and the number of units it buys y, since the two always add up to a constant: $y + z = I^{max} - I^*$. This allows the market to directly infer the acquirer's technological endowments from the target's technology level when the former is unobservable.

Figure 2(b) plots the intrinsic value V as a function of z (blue solid line) under the optimal investment strategy. It increases monotonically in z. Although gap bidders converge to the same final technology level, a smaller initial endowment still reduces intrinsic value due to the higher cost of buying technology units.

Notably, acquisitions create value in this model. The red dotted line shows firms' optimal investment through building when buying is not available (labeled as x^s). Gap bidders must now substitute cheaper bought units with more expensive built ones, resulting in lower final technology levels. This, in turn, reduces intrinsic value, as illustrated by the red dotted line V^s in Figure 2(b). Appendix A.1.2 provides the formal expressions for the optimal investment strategies and intrinsic values discussed above.

Overall, the model implies that technological laggards—those with smaller technology endowments—systematically self-select into acquisitions to close technological gaps. While acquisitions allow them to narrow the gap and enhance intrinsic value, these firms nevertheless remain at the lower end of the value distribution.

2.2 Informational frictions and market reactions

In this section, I derive the market reactions experienced by gap bidders subject to informational frictions. Specifically, I make the following assumptions.

1. The market observes a firm's initial technological endowment z with probability $p_A \in (0, 1)$. With probability $1 - p_A$, it instead observes a noisy signal drawn from a distribution with cumulative distribution function (CDF) F_Z, reflecting the cross-sectional distribution of z.

2. The market does not observe x, the number of technology units built by the firm.

3. For gap bidders, the market observes y, the number of technology units bought by the firm (from the target firm), with probability $p_T \in (0, 1)$.

4. Firms undertake acquisitions for other purposes. Their NPV is normalized to zero for simplicity. When y is unobserved, the market treats gap-closing acquisitions as indistinguishable from other-purpose acquisitions.

These first three assumptions are consistent with real-world observations that ongoing technological development is highly opaque, whereas established technology is only partially

observable. The parameters p_A and p_T capture the degree of transparency of the acquirer and the target, respectively. The last assumption underscores the difficulty of ascertaining the underlying motives for acquisitions.

Following the assumptions above, the pre-announcement market value of a gap bidder with endowment z can be expressed as a weighted average between its intrinsic value and the prior market value across all firms $\overline{V}$,

$$V_0(z) = p_A \cdot V(z) + (1 - p_A) \cdot \overline{V}, \tag{11}$$

where the subscript 0 denotes that the value is the expectation conditional on the pre-announcement information set. The function $V_0(z)$ is increasing in z.

At an acquisition announcement, a gap bidder reveals y with probability p_T. As discussed earlier, revealing y is equivalent to revealing z. Therefore, the acquisition announcement allows the market to observe the acquirer's endowment z more precisely, with posterior probability

$$p_{AT} = 1 - (1 - p_A)(1 - p_T), \tag{12}$$

which is greater than p_A. Notably, this expression assumes that the market's ability to observe z and its ability to observe y are independent. This assumption may be violated if market participants rely on overlapping information sets or are subject to common biases. Although the exact specification does not impact the findings in this section, the next section relaxes this assumption.

The post-announcement market value of a gap bidder can be expressed as a weighted average of three components: its intrinsic value $V(z)$, the prior market value across gap bidders $\overline{V}_G$, and the prior market value across all firms $\overline{V}$:

$$V_1(z) = p_{AT} \cdot V(z) + (1 - p_{AT}) \cdot \left[p_T \cdot \overline{V}_G + (1 - p_T) \cdot \overline{V} \right], \tag{13}$$

where the subscript 1 denotes that the value is the expectation conditional on the post-announcement information set. The function $V_1(z)$ is increasing in z. Notably, $\overline{V}_G < \overline{V}$, since gap bidders tend to have lower intrinsic value than non-gap bidders. For notational convenience, define

$$\tilde{V}_G \equiv p_T \cdot \overline{V}_G + (1 - p_T) \cdot \overline{V}. \tag{14}$$

It is straightforward to show that $\tilde{V}_G < \overline{V}$.

The announcement return—empirically measured as the abnormal return (AR)—can then be expressed as the relative change in the acquirer's market value around the acquisition

announcement:

$$AR(z) = \frac{V_1(z) - V_0(z)}{V_0(z)}. \tag{15}$$

An acquisition announcement conveys two distinct signals. First, the decision to acquire implies that the acquirer might be a technological laggard. Second, the choice of target provides additional information about the acquirer's technological standing. Accordingly, $AR(z)$ can be decomposed into these two components:

$$AR(z) = \underbrace{(1 - p_A) \cdot \frac{\tilde{V}_G - \overline{V}}{V_0(z)}}_{\text{driven by the announcement}} + \underbrace{(p_{AT} - p_A) \cdot \frac{V(z) - \tilde{V}_G}{V_0(z)}}_{\text{driven by the target choice}}. \tag{16}$$

The first component, $\tilde{V}_G - \overline{V}$, captures the shift in the market's prior beliefs about the firm's intrinsic value upon the announcement: with probability p_T, the market now recognizes it as a gap bidder. The scaling factor $1-p_A$ indicates that this component is more pronounced when the acquirer is opaque—i.e., when p_A is low, and when the market relies more heavily on prior beliefs than on firm-specific information.

The second component, $V(z) - \tilde{V}_G$, captures the market's further updating of a gap bidder's value upon learning its true technological position. The scaling factor $p_{AT} - p_A$ reflects the incremental precision with which the market observes the gap bidder's technological standing. It implies that the target-driven component is more pronounced when the acquirer is opaque (low p_A) and when the target is transparent (high p_T, and thus high p_{AT}).

The second component predicts substantial cross-sectional variation in acquirer returns driven by firm characteristics. It therefore constitutes a central focus of the empirical analysis in this paper.

2.3 Model predictions and hypotheses

This section derives testable predictions from the model and formulates the corresponding hypotheses.

2.3.1 The level of acquirer returns

Given that gap bidders tend to be technological laggards, an average gap bidder should experience a negative announcement return. The announcement return for an average gap

bidder can be expressed as

$$\overline{AR} = \frac{\overline{V}_1 - \overline{V}_0}{\overline{V}_0} = (1 - p_A) \cdot p_T \cdot \frac{\overline{V}_G - \overline{V}}{\overline{V}_0}, \tag{17}$$

where $\overline{V}_0$ and $\overline{V}_1$ denote the average pre- and post-announcement market value across all gap bidders, respectively. Since $\overline{V}_G < \overline{V}$, $\overline{AR}$ is negative, in line with the intuition. This implies that cost-driven acquisition decisions by gap bidders can generate systematically negative acquirer returns, even when the acquisitions are value-enhancing.

Notably, the announcement return converges to zero as the target becomes more opaque (i.e., as p_T converges to zero), since greater opacity makes it increasingly difficult to ascertain the gap-closing motive underlying the acquisition. This prediction is consistent with empirical evidence that acquirer announcement returns are higher in private-target deals than in public-target deals, as private targets are more opaque.

2.3.2 The cross-section of acquirer returns

Since the target's technology complements that of the acquirer, a larger y—the number of technology units acquired from the target—implies a weaker technological standing of the acquirer and, therefore, more negative announcement AR. Given $z + y = I^{max} - I^*$ for gap bidders, differentiating $AR(z)$ with respect to y yields

$$\begin{aligned} &d_y AR(z) = -\Phi \cdot c_y, \\ &\text{where } \Phi = \frac{(p_{AT} - p_A \cdot p_{AT}) \cdot \overline{V} - (p_A - p_A \cdot p_{AT}) \cdot \tilde{V}_G}{[V_0(z)]^2}. \end{aligned} \tag{18}$$

Given that $p_{AT} > p_A$ and $\overline{V} > \tilde{V}_G$, it follows that $\Phi > 0$. Therefore, a gap bidder's announcement return decreases with the number of the target's technology units y. Consequently, the target's technology serves as a negative signal of the acquirer's technological standing.

Crucially, the model's prediction only applies when the target's technology lies within the current technology set of the acquirer's industry—i.e., when the technology represented by y is a subset of that represented by z^{max}. It does not extend to cases where the target firm's technology is novel to the industry's existing technology, which are treated as other-purpose acquisitions. Accordingly, a larger y effectively captures a greater *overlap* between the target's technology and the technology set of the acquirer's industry, implying a larger technological gap for the acquirer and, consequently, leading to a more negative market reaction. This observation underpins the empirical measure used in the paper.

Notably, the derivative is proportional to the MC of acquisition, $c_y = p + f + \beta$. The

intuition is that, although gap bidders ultimately converge to the same technology level, those with weaker technological standing incur higher costs when acquiring technology. This cost differential, in turn, drives cross-sectional variation in announcement ARs.

Moreover, Equation (18) does not capture the full impact of revaluations. The revelation of a gap bidder's technological standing updates the market's assessment of its value in two steps. First, the market infers the bidder's weak technological position and revises its value downward. Second, this revealed weakness rationalizes the choice of target. This second step generates unexpected synergies that partially offset the downward revaluation. Accordingly, Equation (18) captures the *net* revaluation effect as a function of y. Section A.1.3 in the Appendix provides a formal decomposition of announcement AR into revaluation and unexpected synergy components.

Taken together, the model predicts that the cost-driven acquisition decision results in announcement returns that are systematically negative for an average gap bidder, and are more negative when the target's technology overlaps more with that of the acquirer's industry. This leads to the first hypothesis of the paper:

Hypothesis 1. *Acquirer returns are negatively correlated with the overlap between the target's technology and that of the acquirer's industry.*

2.3.3 Firm opacity as a moderator

Earlier analyses show that a gap bidder's announcement return includes a component driven by the target choice, which is more pronounced when the opacity difference between the acquirer and the target widens (see Equation (16)). This implies that $d_y AR(z)$, the marginal effect of y on $AR(z)$ (Equation (18)), should be amplified as the acquirer-target opacity difference increases.

Define the opacity difference as

$$\Delta \equiv p_T - p_A. \tag{19}$$

The red line in Figure 3 visualizes the shape of $d_y AR(z)$ as a function of Δ. The marginal effect $d_y AR(z)$ is negative except when the acquirer's technology is fully observable (leftmost part of the figure) and becomes more negative as Δ increases. This leads to the second hypothesis of the paper:

Hypothesis 2. *The negative relationship between acquirer returns and the technology overlap between the target and the acquirer's industry is moderated by the relative opacity between the acquirer and the target. A more transparent target or a more opaque acquirer—hence a larger opacity difference between the acquirer and the target—amplifies the negative relationship.*

This prediction, however, assumes that the information the market observes about the acquirer's and the target's technology is independent (the posterior probability of observing the acquirer's true technological standing follows Equation (12)). In reality, the two pieces of information may overlap. To illustrate this point, suppose that the acquirer's technology z and the target's technology y each consist of two components: public information (denoted z_1 and y_1, e.g., patents) and confidential information (denoted z_2 and y_2, e.g., unpatented proprietary inventions), such that $z = z_1 + z_2$ and $y = y_1 + y_2$. The market can only infer the acquirer's technology within a given component from the target's technology in the corresponding component.

Consequently, if an opaque acquirer reveals only public information, the target must be strictly more transparent (i.e., reveal information beyond the public component) to shed additional light on the acquirer's technological standing. Otherwise, the target's information is *nested* within the market's existing knowledge about the acquirer and does not lead to incremental learning about z.

The blue solid line in Figure 3 visualizes $d_y AR(z)$ under this assumption. Although the function remains negative throughout, the slope is nearly flat when $\Delta < 0$. In this region, the target does not provide incremental information about the acquirer's unobserved technological component, so belief updating about the acquirer's technological standing is muted. A more transparent target still helps the market better infer the acquirer's intention to close the gap (hence the downward slope), but not the magnitude of the gap itself.

By contrast, the slope becomes substantially steeper when $\Delta > 0$. In this region, the novel information revealed by the target enables a sharper update of the market's belief about the acquirer's technological standing. As a result, the a more transparent target simultaneously sharpens the inference about both the acquirer's technological gap and its gap-closing motive, leading to a stronger negative marginal effect on announcement returns. Section A.1.4 in the Appendix provides a proof of the numerical relationships above.

This leads to two refined hypotheses in addition to Hypothesis 2:

Hypothesis 2a. *The negative relationship between acquirer returns and the technology overlap between the target and the acquirer's industry does not exhibit a discontinuity at the point where the target becomes more transparent than the acquirer.*

Hypothesis 2b. *The negative relationship between acquirer returns and the technology overlap between the target and the acquirer's industry exhibits a discontinuity at the point where the target becomes more transparent than the acquirer.*

2.3.4 Announcement returns for peers

An acquisition announcement enables the market to better distinguish a technological laggard from other firms. When the number of firms in the industry is finite, the announcement may simultaneously trigger revaluations of other firms in the opposite direction.

To illustrate this point, suppose there are N firms in the market, with $p_A^i = p_A \ \forall i \in \{1, 2, ..., N\}$. Suppose gap bidder g with $z = z_g$ makes an acquisition announcement. With a probability of p_{AT}, it reveals z_g; with a probability of $1 - p_{AT}$, the market updates its prior market value to $\tilde{V}_G$.

Accordingly, the market updates its prior belief on the other firms' intrinsic value:

$$\overline{V}^{\neg g} = p_{AT} \cdot \frac{N \cdot \overline{V} - V(z_g)}{N-1} + (1 - p_{AT}) \cdot p_T \cdot \frac{N \cdot \overline{V} - \tilde{V}_G}{N-1} = \frac{N \cdot \overline{V} - V_1(z_g)}{N-1}. \tag{20}$$

Accordingly, their post-announcement value can be expressed as

$$V_1^{\neg g}(z_i) = p_A \cdot V(z_i) + (1 - p_A) \cdot \overline{V}^{\neg g}. \tag{21}$$

Two observations follow. First, the aggregate market value across firms remains unchanged after the announcement:

$$V_0^{total} = \sum_{i=1}^{N} V_0(z_i) = N \cdot \overline{V}, \tag{22}$$

$$V_1^{total} = \sum_{i \neq g}^{N} V_1^{\neg g}(z_i) + V_1(z_g) = N \cdot \overline{V}. \tag{23}$$

The intuition is that the market has already priced in all value-relevant information at the industry level ex ante, including acquisition synergies. Thus, acquisition announcements per se do not create or destroy value, but instead redistribute it across firms.

Second, given that aggregate market value remains constant, the announcement returns experienced by gap bidders must be offset by price movements in the opposite direction for other firms. The announcement return for other firms is given by

$$AR^{\neg g}(z_i) = \frac{V_1^{\neg g}(z_i) - V_0(z_i)}{V_0(z_i)} = \frac{1}{N-1} \cdot \frac{\overline{V} - V_1(z_g)}{V_0(z_i)}, \tag{24}$$

which is decreasing in z_g and, therefore, increasing in y_g—the number of technology units that gap bidder g acquires from the target.

Taken together, acquisition announcements enable the market to better overcome informational frictions and distinguish between technological laggards and leaders, thereby driving

market values closer to their intrinsic levels. This leads to the third hypothesis of the paper:

Hypothesis 3. *Upon a gap bidder's acquisition announcement, its peers experience changes in their market value. These returns are positively correlated with the degree of technological overlap between the target and the acquirer's industry.*

3 Data and methodology

In this section, I introduce the acquisition sample analyzed in the paper and the data sources employed. I then outline the empirical strategy, detailing how firms' technological profiles are quantified and how the measure of the revealed technological gap is constructed. Finally, I present summary statistics to further illustrate the sample.

3.1 Sample construction

To test the hypotheses, I construct a sample of announced acquisitions involving publicly listed acquirers between 1990 and 2020, using data from SDC via Wharton Research Data Services (WRDS). I supplement this sample with trading data from CRSP, financial data from COMPUSTAT, and patent data from the USTPO. Table A1 summarizes the sample selection process, including the filters applied and the number of observations remaining at each step. The final sample includes 1,044 public-target acquisitions and 962 private-target acquisitions.

Notably, I exclude acquisitions in which the target is a subsidiary, as I cannot reliably identify which patents filed under the parent company are transferred with the subsidiary. This exclusion is important because the analysis relies on patent portfolios to quantify firms' technological profiles.

To reflect industry-wide technological dynamics, I compile a broader sample of public and private companies by combining the universes of COMPUSTAT and CapitalIQ companies. Supplemented with hand-collected Standard Industrial Classification (SIC) codes for CapitalIQ firms, the final sample consists of 235,376 distinct companies, both public and private.

To characterize the technological profiles of all sample firms, I implement a frequency-based name-matching algorithm following Hall, Jaffe, and Trajtenberg (2001) to link patents to firms.[6] In total, I successfully link approximately 3 million patents to companies in the

[6]Further technical details are provided in Appendix A.2.

combined sample.[7] Of the linked patents, approximately 63% are contributed by public companies, 35% by private firms, and the remainder by institutions such as investment funds and universities. Overall, the linked patents account for about 40% of all patents filed globally by companies, individuals, and other entities between 1980 and 2020 with the USTPO, ensuring the representativeness of the patent sample.

It is noteworthy that only a subset of firms actively engage in patenting activities. As a result, the final sample is best understood as one comprising technology-oriented acquisitions.

3.2 Research design

The goal of the empirical strategy is to quantify the technological overlap between the target and the acquirer's industry—the proxy for the acquirer's revealed technological gap, and to examine how this revealed-gap measure correlates with the stock market reactions to acquisition announcements. I first quantify firms' technological profiles leveraging patent data. Then, I construct a measure for the acquirer's revealed technological gap. Next, I describe the dependent variables—i.e., cumulative abnormal returns (CARs) around acquisition announcements, proxies for corporate opacity, and control variables. Lastly, I present and discuss descriptive statistics of the sample.

3.2.1 Quantifying technological profiles

To characterize a firm's technological profile, I rely on its patent portfolio. Patents are a valuable source of information for two main reasons. First, to be granted a patent, an invention must demonstrate real-world applicability, novelty over all prior art, and a non-obvious contribution beyond the knowledge of a skilled practitioner in the field. In this sense, patented inventions capture the technological frontier at the time of filing. Second, patents confer temporary monopolistic rights (typically lasting up to ten years) that allow the firm to exclusively commercialize the protected inventions. These rights often translate into substantial economic value. Indeed, a rich literature has documented patents' value effect.[8]

Patent classification codes are commonly used by researchers to identify technological fields.[9] In this paper, I employ the Cooperative Patent Classification (CPC) codes for this

[7] There are two types of patents: utility patents and design patents. According to the USTPO, a utility patent protects the way an article is used and works (35 U.S.C. §101), while a design patent protects the way an article looks (35 U.S.C. §171). This paper focuses exclusively on utility patents, as they better reflect the technological content of innovation.

[8] See, for example, Farre-Mensa, Hegde, and Ljungqvist (2020), Hall, Thoma, and Torrisi (2007), Kogan et al. (2017), and Pakes (1985).

[9] See, for example, Hall, Jaffe, and Trajtenberg (2001), Hegde, Herkenhoff, and Zhu (2023), and Seru

purpose. The CPC scheme follows a hierarchical structure that becomes increasingly granular at lower levels. I focus on the *class* level, which groups patents into 137 distinct technological classes.[10] For robustness, I also replicate the main analysis using the *subclass* level, comprising 680 categories. The results remain both qualitatively and quantitatively similar.

For each firm i in year t, I extract all granted patents filed by the firm over the preceding 10 years (from year $t-1$ through year $t-10$).[11] Then, I count the number of patents that fall into each of the 137 patent classes and construct a 137-dimensional vector $\mathbf{P}_{it} = (p_{it}^1, p_{it}^2, \ldots, p_{it}^{137})$, where each element p_{it}^j represents the number of patents in class j.[12] Additionally, I denote the patent vectors for acquirers and targets as $\mathbf{P}_{it}^{Acq}$ and $\mathbf{P}_{it}^{Tar}$, respectively. The $\mathbf{P}$-vector describes the distribution of a firm's past patenting activity across technological classes and serves as a proxy for the firm's technological orientation.

Furthermore, for each acquirer, I characterize the technology set of its industry as of the year preceding the acquisition announcement. To do so, I aggregate the $\mathbf{P}$-vectors across all firms operating in the same two-digit SIC industry as the acquirer. To avoid endogeneity, I exclude both the acquirer and the target—if the target operates in the same two-digit SIC industry—from this calculation. The resulting vector that defines the industry technology set for acquirer i in year t is

$$\mathbf{P}_{it}^{AInd} = \sum_{k \in S_i} \mathbf{P}_{kt}, \tag{25}$$

where S_i denotes the set of firms in acquirer i's 2-digit SIC industry, excluding the acquirer and, if applicable, the target.

There exist alternative ways to characterize technological profiles. To ensure that the findings of this paper are not specific to the chosen empirical approach, Section 5.3 implements four alternative definitions: (1) a more granular patent classification scheme, (2) shorter patenting histories (5 years), (3) a weighting scheme based on the citations patents receive, and (4) a weighting scheme based on patents' economic value, inferred from stock market reactions to patent news. The main findings hold across these alternative definitions.

3.2.2 Measuring revealed technological gaps

Acquirers' technological gaps are not directly observable. As established in Section 2, the market can infer an acquirer's technological gap based on the technological profile of its ac-

(2014).

[10] The CPC scheme evolves as new technologies emerge. All CPC codes used in this paper reflect the version as of the end of 2023 and are applied retroactively to earlier patents.

[11] I also devise an alternative definition considering 5-year patenting histories. The main results of the paper are robust to the choice of this window. See Section 5.3 for details.

[12] I also devise an alternative definition based on 680 patent "subclasses". The main results of the paper are robust to the granularity of technological fields. See Section 5.3 for details.

quisition target. Following this logic, I capture the acquirer's revealed technological gap as the technological overlap between the target and the acquirer's industry. Greater overlap indicates that the acquirer's technological needs lie more squarely within its industry's existing technologies, hence signaling its weak technological standing. I label the resulting measure as the *GapSignal* score.

Leveraging patent-based measures, I define the *GapSignal* score as the cosine similarity between the target's patent vector $\mathbf{P}_{it}^{Tar}$ and the acquirer's industry set $\mathbf{P}_{it}^{AInd}$:

$$GapSignal_{it} = \frac{\mathbf{P}_{it}^{Tar} \cdot \mathbf{P}_{it}^{AInd}}{\|\mathbf{P}_{it}^{Tar}\| \cdot \|\mathbf{P}_{it}^{AInd}\|}, \tag{26}$$

where $\|\cdot\|$ denotes the Euclidean norm of a vector. Cosine similarity captures the angular alignment between two vectors while ignoring their magnitude, and thus reflects technological orientations rather than overall patenting intensity. This choice is grounded in the fact that the number of patents a firm holds may reflect strategic patenting behavior, rather than its technological capabilities. For instance, firms may file patents defensively or preemptively in response to competitive pressure (Bessen, 2003; Cappelli et al., 2023; Gurgula, 2020). As a result, patents may better capture a firm's technological mix than its technological strength (Reeb & Zhao, 2020). Cosine similarity captures this compositional element while abstracting away from differences in patent volumes.

In addition, to control for the acquirer's observable technological position relative to the industry technology set, I construct an *Alignment* score as the cosine similarity between the acquirer's patent vector $\mathbf{P}_{it}^{Acq}$ and its industry technology set $\mathbf{P}_{it}^{AInd}$:

$$Alignment_{it} = \frac{\mathbf{P}_{it}^{Acq} \cdot \mathbf{P}_{it}^{AInd}}{\|\mathbf{P}_{it}^{Acq}\| \cdot \|\mathbf{P}_{it}^{AInd}\|}. \tag{27}$$

This score reflects the technological overlap between the acquirer and its industry. Higher values indicate better alignment, while lower values suggest a greater observable gap. *Alignment* captures public information observable prior to the acquisition announcement. In contrast, *GapSignal* captures information specific to the acquirer-target match and becomes available only upon the announcement of the acquisition.

3.2.3 Other variables

Cumulative abnormal returns (CARs). The main dependent variable in this paper is CARs around acquisition announcement dates. I compute CARs for both acquirers and targets using the four-factor model of Fama and French (1993) and Carhart (1997). The model is estimated over a 200-trading day estimation window that ends 20 trading days prior

to the announcement to avoid event contamination. CARs are calculated over two symmetric event windows: $[-1, 1]$ and $[-3, 3]$, where day 0 is the announcement date, or the immediate trading day after it.

Corporate opacity measures. To empirically examine the moderating role played by acquirer and target opacity, I employ two widely adopted proxies for firm-level opacity: bid-ask spreads and analyst following.[13] I follow Abdi and Ranaldo (2017) to estimate effective bid-ask spreads using closing, high, and low prices, as closing bid-ask quotes reported by CRSP do not fully span my sample period. Analyst following is measured as the number of analysts providing earnings forecasts for a firm within a given calendar year, based on I/B/E/S data.

Firm-level control variables. For each pair of acquirer and target, I construct the following variables for both firms: total assets, patent stock, Tobin's Q, and market leverage. These variables capture, respectively, firm size, historical patenting activity, market valuation, and financial condition. The first three firm-level controls are measured in logarithmic form. All four controls are based on the most recent fiscal year preceding the acquisition announcement. Detailed definitions are provided in Table A2.

Deal-level control variables. I characterize each acquisition using the following dimensions: the percentage of the transaction paid in cash (*% in cash*); whether the deal is structured as a tender offer (*Tender offer*); whether the acquisition is hostile (*Hostile*); the presence of competing bids (*Competitive*); the acquirer's toehold, i.e., pre-existing ownership stake in the target (*Toehold*); whether the deal is cross-border (*Cross-border*); whether the acquirer and target operate in the same industry (*Horizontal*) or are product market rivals (*Prod. market rival*), based on the classification by Hoberg and Phillips (2010); and whether the deal was preceded by public rumors (*Rumored deal*). These variables help account for deal structure, strategic complexity, and competitive environment, all of which may influence market reactions independent of technological considerations. Detailed definitions are provided in Table A2.

[13]For example, Anderson, Duru, and Reeb (2009) combine both proxies into a composite opacity index. Daske et al. (2008), Cheng, Courtenay, and Krishnamurti (2006), and Leuz and Verrecchia (2000) show that improved disclosure reduces bid-ask spreads. Hong, Lim, and Stein (2000) find that analyst following accelerates information diffusion.

3.3 Descriptive statistics

Figure 4 breaks down the sample by year. Panel A shows the annual distribution of acquisitions in the sample from 1990 to 2020, separated by the target's listing status. The volume of technology-oriented acquisitions rises sharply during the late 1990s and early 2000s, coinciding with the dot-com bubble, and shows another uptick in the mid-2010s, potentially reflecting renewed interest in digital transformation and platform-based business models.

Panel B displays the average acquirer CARs over two symmetric event windows: CAR(-1,1) and CAR(-3,3). Acquirer returns around public-target announcements are often, on average, negative over the sample period, whereas private-target acquisitions are almost consistently associated with positive acquirer returns. This divergence is persistent across time and event windows.

Panel C plots the average *GapSignal* score by year. While both public- and private-target deals show an upward trend, the average *GapSignal* is consistently higher for public-target acquisitions, suggesting that public targets tend to be more technologically aligned with the technology set of the acquirer's industry.

Table 1 presents summary statistics for the sample of acquisitions, separated by whether the target is a public or private firm. Panel A reports acquirer characteristics. Consistent with prior literature and the time-series pattern shown in Figure 4, acquirers of public targets experience significantly negative announcement CARs (mean CAR(-1,1) = -1.2%, $p < 0.01$; mean CAR(-3,3) = -1.6%, $p < 0.01$), while acquirers of private targets earn significantly positive CARs (mean CAR(-1,1) = 2.3%, $p < 0.01$; mean CAR(-3,3) = 1.7%, $p < 0.01$). Public-target acquirers are substantially larger, hold larger patent portfolios, and have slightly higher market leverage. All differences are statistically significant at the 1% level.

Panel B compares target characteristics. In line with established findings, target firms experience large positive announcement returns (mean CAR(-1,1) = 24.9%, $p < 0.01$; mean CAR(-3,3) = 25.5%, $p < 0.01$). Public targets are significantly larger and more patent-intensive than private targets, with all differences highly significant.

Panel C summarizes deal-level features. Public-target acquisitions exhibit significantly higher *GapSignal* scores, consistent with Figure 4. By contrast, the *Alignment* score, which measures the acquirer's technological overlap with its industry, does not differ significantly between the two groups. Public-target deals are also more likely to be structured as tender offers, hostile, cross-border, or rumored, whereas private-target acquisitions are more likely to result in deal completion. All these differences are statistically significant. Finally, public-target acquisitions involve significantly higher cash consideration, which may reflect acquirers' stronger bargaining position when negotiating with private firms.

Lastly, the sample acquirers are concentrated in a few technology-intensive industries. The

five most represented industries are business services (SIC 73; 17.6%), precision equipment (SIC 38; 16.4%), chemicals (SIC 28; 14.9%), electronics (SIC 36; 14.7%), and machinery (SIC 35; 11.2%). Together, these sectors account for approximately 75% of the sample. The remaining acquirers are distributed across other manufacturing industries (11.8%) and service sectors (13.6%). This distribution underscores the sample's strong orientation toward R&D- and innovation-driven firms.

4 Analysis

This section presents the main empirical analyses. I first document a negative relationship between acquirer returns and my measure of the revealed technological gap among acquirers of public targets, while no such relationship is observed for acquirers of private targets. I then examine the role of informational frictions in shaping market reactions to acquisition announcements and show that acquirers are negatively revalued only when their targets are more transparent than they are. Building on this insight, I demonstrate that transparent private targets likewise trigger negative revaluations of their acquirers. Finally, I relate my findings to prior studies and show that the effect is not captured by previously documented acquisition antecedents.

4.1 Technological gaps and acquirer revaluation

Figure 5 sorts public- and private-target acquisitions into terciles based on the *GapSignal* score and reports the average acquirer CARs within each tercile for two event windows: CAR(-1,1) and CAR(-3,3). Among public-target acquisitions (Panel A), a clear negative relationship emerges between *GapSignal* and acquirer returns, consistent with Hypothesis 1. This pattern suggests that when a public target overlaps with the acquirer's industry technologically, the market interprets the deal as revealing a larger technological gap for the acquirer, prompting negative revaluations. Among private-target acquisitions (Panel B), a similar negative association is also evident, although average CARs remain positive across all terciles.

Next, I formally test the relationship between acquirer returns and the revealed technological gap by estimating the following regression model:

$$CAR(-\tau,\tau)_{id} = \gamma \cdot GapSignal_d + \mathbf{X}_{id} \cdot \mathbf{\Gamma} + \alpha_u + \alpha_v + \alpha_t + \varepsilon_{id}, \tag{28}$$

where i indexes acquirers, d indexes acquisition deals, u and v denote 2-digit SIC industry of the acquirer and the target, respectively, and t represents the calendar year. $\mathbf{X}_{id}$ is a vector of

control variables capturing firm- and deal-level characteristics, as introduced in Section 3.2.3. The regression includes industry and year fixed effects α_u, α_v, α_t to account for unobserved heterogeneity across industries and time. The coefficient of interest, γ, captures the marginal effect of the revealed technological gap, proxied by *GapSignal*, on the acquirer's abnormal return.

4.1.1 Acquirers of public targets

I start with the sample of acquirers of public targets. Table 2 reports the regression results based on Equation (28), using public-target acquirer's CAR(-1,1) as the dependent variable in Panel A, while CAR(-3,3) in Panel B.

Across all specifications in both panels, the coefficient on *GapSignal* is negative and statistically significant at the 1% level, except in Column (3) Panel A, where it remains significant at the 5% level. This result supports the baseline hypothesis: when a target overlaps the acquirer's industry technologically to a greater extent, the market interprets the acquisition as revealing a larger underlying technological gap in the acquirer, triggering a negative revaluation.

Notably, the more stringent fixed effects introduced in Columns (4) and (5) of both panels strengthen the estimated relationship. Column (4) controls for time-varying factors specific to either the acquirer's or the target's industry by including acquirer-year and target-year fixed effects. Column (5) further saturates the specification by controlling for time-varying factors at the acquirer-target industry-pair level. Importantly, these saturated fixed effects do not weaken the coefficient on *GapSignal*; rather, they amplify it substantially. Relative to Columns (1) to (3), the coefficient magnitude roughly doubles in Panel A and increases by over 50% in Panel B. This pattern suggests that the fixed effects absorb considerable noise and mitigate attenuation bias, thereby sharpening the estimated effect of the revealed technological gap on acquirer returns.

The magnitude of the effect is also economically meaningful. A one-standard deviation increase in *GapSignal* is associated with a 0.83-1.79 percentage point decline in CAR(-1,1) and a 1.23-1.83 percentage point decline in CAR(-3,3). These changes represent approximately 10-21% of the standard deviation of the respective CARs.

Control variables. The effect of *Alignment*, which captures the acquirer's own technological overlap with its industry, is negative but statistically significant only in Column (2) of Panel A. The effect becomes insignificant once firm-level controls are introduced in Column (3), suggesting that the market has already priced in the acquirer's observable technological orientation prior to the acquisition announcement.

A higher share of cash payment (*% in cash*) is consistently associated with more positive announcement returns. This finding aligns with prior literature, which interprets cash financing as a signal of acquirer undervaluation. The economic magnitude is comparable to—even slightly smaller than—that of *GapSignal*: a one-standard deviation increase in the cash share corresponds to an 8-16% standard deviation increase in CARs. This underscores the strong explanatory power of the informational mechanism captured by *GapSignal*.

Additionally, larger acquirers tend to earn significantly higher announcement returns, whereas acquisitions of larger targets are associated with lower acquirer returns, consistent with established findings in the literature.[14] This pattern may reflect relative bargaining power in negotiations or the greater complexity and integration costs associated with acquiring large targets. Other untabulated control variables do not exhibit consistent or robust effects across specifications.

Overall, these results reaffirm the visual pattern observed in Figure 5, Panel A, and provide strong empirical support for Hypothesis 1.

4.1.2 What if the gap is known?

Before proceeding to the acquirers of private targets, this section digresses to examine a central assumption of the paper: that the market cannot fully observe all acquirers' technological standing prior to the acquisition announcement. If the acquirer's technological standing were common knowledge ex ante, acquisition announcements could not reveal new information about it, thereby calling my interpretation of the findings into question.

While it is challenging to directly substantiate what the market does not know, it is easier to identify cases in which the market arguably has greater knowledge of the acquirer's technological basis prior to the acquisition announcement. In such cases, the market may be able to partially infer the acquirer's gap-closing intention ex ante, thereby weakening the effect documented above.

The literature has established that toehold stakes increase the probability of a full acquisition by the toehold holder. Akhigbe, Martin, and Whyte (2007) show that the size of the toehold is positively associated with the likelihood of acquisition. Therefore, gap-closing acquisitions by acquirers with larger toehold stakes may reveal less about the acquirer's technological basis at the time of the acquisition announcement, as the market may be able to partially infer this information ex ante.

Inspired by the argument above, I re-estimate Equation (28) while interacting *GapSignal* with *Toehold*, a continuous variable capturing the acquirer's percentage pre-bid ownership in the target. Table 3, Panel A, reports the results. The coefficients on *GapSignal* are con-

[14] See, for example, Loderer and Martin (1990) and Moeller, Schlingemann, and Stulz (2004).

sistently negative and significant across all specifications, in line with the findings in Table 2. Notably, the coefficients on the interaction term are positive throughout and significant in four of the six columns. These results indicate that a larger toehold stake is associated with a less negative marginal effect of *GapSignal* on acquirer returns, consistent with the interpretation that toehold stakes reveal information about the acquirer's technological standing prior to the acquisition announcement.

Second, I consider rumored deals. Like toehold stakes, rumors partially reveal information that would otherwise become public at the time of the acquisition announcement. Consequently, I expect rumors to weaken the effect documented earlier. Table 3, Panel B, reports the results. Similar to the case with toehold stakes, the interaction term between *GapSignal* and *Rumored deal*—an indicator equal to one if a deal is marked as rumored by SDC—yields positive coefficients throughout, indicating that rumors weaken the negative correlation between *GapSignal* and acquirer returns. However, the statistical significance is weaker, likely because rumors carry less weight than toehold ownership and therefore lend less credibility to the inferences drawn from them.

Lastly, I consider back-to-back acquisitions. To the extent that preceding acquisitions may have already partially revealed an acquirer's technological standing, I expect the market to find subsequent acquisitions less surprising. Table 3, Panel C, reports the results. *Back-to-back* is an indicator equal to one if an acquisition is preceded by another within 50 calendar days. The coefficients on the interaction term are again positive across the board, with four of the six also statistically significant. These results further corroborate the findings in Panels A and B and support the interpretation that greater ex ante market knowledge of the acquirer's technological standing leads to less revelation upon the acquisition announcement.

Notably, the coefficients on *Toehold*, *Rumored deal*, and *Back-to-back* in Panels A, B, and C, respectively, are almost unanimously negative, albeit with varying levels of statistical significance. This pattern is consistent with prior studies documenting declining acquirer returns to serial acquisitions (Aktas, de Bodt, & Roll, 2011; Fuller, Netter, & Stegemoller, 2002) and lower acquirer returns for anticipated deals Tunyi (2021). In contrast, my findings highlight a novel channel through which anticipation and serial acquisitions may increase acquirer returns by redistributing information revelation over time.

Taken together, these results reveal cross-sectional heterogeneity in the market's knowledge of the acquirer's pre-announcement technological standing, thereby indirectly supporting the assumption that at least some aspects of the acquirer's technological standing are not common knowledge prior to the acquisition announcement. In this sense, these findings validate the paper's theoretical and empirical premise.

4.1.3 Acquirers of private targets

Table 4 presents regression results using the same specification as in Table 2, but based on the sample of private-target acquisitions.[15]

In contrast to the results for public-target acquirers, the coefficient on *GapSignal* is statistically indistinguishable from zero across all specifications in both panels. This finding stands in contrast to the suggestive pattern in Figure 5, Panel B, and indicates that the market does not systematically revalue acquirers of private targets based on the technological profile of the latter.

While initially puzzling, this finding is not necessarily inconsistent with the revaluation mechanism. As shown theoretically in Section 2.3.3, the intensity of market revaluations depends critically on the informativeness of the target relative to the acquirer. When the target is more opaque—meaning that the signal it provides about the acquirer's technological position is noisier than the acquirer's own observable characteristics—the revaluation effect can be substantially weakened, or even fully muted in the case of perfectly correlated noise components.

In the context of private-target acquisitions, this condition is likely to hold. First, private firms are not subject to the same regulatory disclosure requirements as their publicly traded counterparts, making them inherently less transparent. Second, as reported in Table 1, private targets in the sample are, on average, less than one-fifteenth the size of their acquirers, and smaller than one-thirtieth the size of public targets. These stark differences in size suggest that private targets are substantially more opaque, which limits the market's ability to draw inferences about the acquirer based on the target's technological profile. As a result, the revaluation effect might be weakened or absent following acquisitions of private firms.

Control variables. The effect of *Alignment* is negative but statistically significant only in Columns (2) and (3) in both panels. Its effect becomes insignificant once time-varying industry heterogeneity is controlled for in Columns (4) and (5), echoing the similar pattern in Table 2.

Interestingly, relative to Table 2, the coefficients on *% in cash* in both panels and acquirer *Log asset* in Panel A reverse in sign, while all other coefficients lose statistical significance and fluctuate around zero across specifications. This underscores the distinct and complex dynamics underlying private-target acquisitions. Lastly, in both panels, the adjusted R^2

[15]Note that some independent variables are not available for private-target acquisitions, including target *Tobin's Q*, target *Market leverage*, and *Prod. market rival*—the indicator for whether the acquirer and the target are product market rivals, since Hoberg and Phillips (2010) only classify product markets for public companies. Additionally, *Competitive* is also excluded because none of the sample private-target acquisitions are subject to competitive bidding.

in Columns (4) to (5) is substantially smaller than in Columns (1) to (3), and even turns negative in Column (4), Panel B. This reflects the fact that the granular fixed effects included in Columns (4) to (5) are oversaturated.

Next, I empirically examine how relative acquirer-target opacity moderates revaluations, and provide further insights into the muted relationship between acquirer returns and revealed technological gaps in private-target acquisitions.

4.2 The moderating role of informational frictions

This section empirically tests Hypothesis 2: a relatively more transparent target strengthens revaluation while a relatively more transparent acquirer dampens it. Section 4.2.1 focuses on acquisitions involving public firms. Section 4.2.2 then turns to acquisitions of private firms.

4.2.1 Opacity and public acquisitions

First, I define the acquirer-target relative opacity as a normalized measure by scaling the difference in the opacity levels of the acquirer and the target, respectively, by their average magnitude:

$$\Omega = \frac{\omega^{Acq} - \omega^{Tar}}{0.5 \cdot (|\omega^{Acq}| + |\omega^{Tar}|)}, \tag{29}$$

where ω denotes a firm's opacity measure. As introduced in Section 3.2.3, I use two standard proxies for opacity: bid-ask spreads and analyst coverage. Firms with wider bid-ask spreads or fewer analysts following are considered more opaque. To ensure consistency in interpretation across proxies, I use the negative of analyst coverage as the opacity measure, so that higher values of ω uniformly indicate greater opacity.

The opacity proxies—bid-ask spreads and analyst following—indicate that 24% and 20% of acquirers, respectively, are more opaque than their targets in public-target acquisitions ($\Omega > 0$). This is not surprising, given that the average acquirer is twice the size of the average target, and the median acquirer is eight times as large as the median target in the public-target acquisition sample, according to Table 1.

To examine how the acquirer-target relative opacity Ω moderates revaluation intensity, i.e., the strength of the relationship between the acquirer's announcement returns and its revealed technological gap. I model the marginal effect of *GapSignal* on acquirer CARs as a function of Ω:

$$\frac{\partial CAR}{\partial GapSignal} = g(\Omega). \tag{30}$$

Figure 3 illustrates possible shapes of the function $g(\cdot)$ predicted by the model. In both cases, the opacity difference between the acquirer and the target amplifies the revaluation

effect. Specifically, the first scenario (red line) assumes that the market's observations of the acquirer and the target are independent. Consequently, the target *always* contributes to the precision with which the market observes the acquirer's technological standing. The second scenario (Equation (52)) assumes that the information provided by the acquirer and the target is nested. Accordingly, the market does not update its belief about the acquirer's technological standing when the target more opaque.

I estimate the function $g(\cdot)$ using two complementary approaches. First, to accommodate the possibility of a cusp at $\Omega = 0$, I model $g(\cdot)$ as a piecewise linear function with a breakpoint at $\Omega = 0$. Second, to allow for more flexible nonlinearity in the $g(\cdot)$, I also adopt a restricted cubic spline (RCS) specification. The RCS approach fits a piecewise cubic polynomial with continuity and smoothness constraints at the knot points, enabling the shape of $g(\cdot)$ to vary smoothly across the domain of Ω. Appendix A.5 introduces the methodology in detail.

Results. Figure 6 plots the estimated function $g(\cdot)$. Panels A and B use bid-ask spreads as the proxy for opacity, while Panels C and D use (negative) analyst following. Panels A and C estimate $g(\cdot)$ using a piecewise linear specification; Panels B and D use a RCS specification.

Across all panels, a consistent pattern emerges: as the target becomes more transparent relative to the acquirer (i.e., as Ω increases), revaluation intensifies. In other words, the marginal effect of *GapSignal* on acquirer returns becomes more negative. This is consistent with the theoretical prediction that the relationship between acquirer returns and the revealed technological gap is negative, and that this effect is amplified when the market can better extract information from the target.

Importantly, the revaluation intensity is close to zero when the target is more opaque than the acquirer (i.e., $\Omega < 0$), suggesting that the market places little weight on the target as an information source in such cases. In contrast, when the target is more transparent (i.e., $\Omega > 0$), the market interprets the target as a more credible source for inferring the acquirer's technological gap, leading to stronger revaluation.

The results also suggest that the negative revaluation of the acquirer is concentrated in roughly one-quarter to one-fifth of public-target acquisitions, while the remaining acquirers experience little to no revaluation. To the extent that the coefficients on *GapSignal* in Table 2 represent the average revaluation intensity across all public-target deals, then acquirers that are more opaque than their targets ($\Omega > 0$) likely experience revaluation effects four to five times stronger than the average, while the rest ($\Omega < 0$) are largely unaffected.

Overall, these results provide strong support for Hypothesis 2 and, specifically, Hypothesis 2b.

4.2.2 Reassessing the private-target puzzle

Having established that negative revaluation in public-target acquisitions concentrates in deals where the target is more transparent than the acquirer, the muted effect observed for private-target acquirers (Table 4) becomes less surprising. Private targets are, on average, significantly more opaque than their public acquirers, which likely limits the market's ability to make inferences based on the target's profile. As a result, the average marginal effect of *GapSignal* is too weak to detect. However, this also suggests a testable implication: if the mechanism holds, I should observe negative revaluation among private-target acquirers when the target is exceptionally transparent.

To this end, I leverage two types of private targets that are arguably more transparent: those that have received late-stage VC funding and those that have issued corporate bonds (or other securities) prior to the acquisition announcement. At the late stage of VC funding, portfolio companies are typically required to reduce information asymmetry with potential investors in preparation for an initial public offering (IPO) or a secondary buyout (SBO). I identify such firms using VC transaction data from Preqin. In total, 276 private targets received VC funding within five years prior to the acquisition announcement, of which 145 obtained at least one late-stage funding round.[16]

Bond issuance by a private firm typically requires adherence to disclosure standards to participate in credit markets. I identify bond issuers using three complementary data sources. First, I identify corporate bond issuances by private firms using records published by the Financial Industry Regulatory Authority (FINRA), which provides direct evidence of credit market offerings. Second, I search for headlines in CapitalIQ that reference corporate bond offerings by private targets, capturing disclosures through financial media. Lastly, I incorporate information from SDC on whether a target has been assigned a SEDOL code prior to the acquisition announcement. Since SEDOL identifiers are typically allocated to firms involved in public or quasi-public securities markets, their presence signals a heightened level of financial visibility. In total, 47 private targets meet these criteria.

Next, I replicate the analyses in Table 4 while interacting *GapSignal*—the acquirer's revealed technological gap—with an indicator for late-stage targets or bond-issuing targets. Table 5 reports the regression results. The interaction term is negative and statistically significant across most specifications (except for Columns (2) and (5) in Panel B), suggesting that acquirers of exceptionally transparent private targets experience negative revaluations when the target overlaps more with the acquirer's industry technologically.

[16] I classify a funding round as late-stage if it is labeled as Growth, PIPE, Pre-IPO, Recapitalization, Secondary Stock Purchase, Series C or later, or Venture Debt.

4.3 Disentangling prediction and surprise

The analyses thus far have implicitly assumed that *GapSignal* captures information that surprises the market. This rests on the premise that, at the time of the acquisition announcement, investors have not fully internalized the acquirer's technological gap. The observed revaluation, then, reflects a market response to newly revealed information conveyed through the acquirer's choice of target. However, this interpretation warrants further scrutiny. A large body of literature has identified a variety of observable firm characteristics—such as size, valuation, growth opportunities, and R&D intensity—that predict acquisition activity. Whether the effect documented here genuinely reflects informational surprises, or whether it simply captures dimensions already priced by the market, remains an open empirical question. Moreover, it is important to assess the extent to which the self-selection mechanism proposed in this paper complements, or overlaps with, existing theoretical frameworks. To this end, I conduct a brief review of relevant studies.

4.3.1 Acquisition antecedents

First, I consider firms' innovation input and output, drawing on Bena and Li (2014), who show that firms with big patent portfolios but low R&D intensity are more likely to become acquirers. Following their definitions, I construct four variables to characterize firms' innovative capabilities and their changes over time: $R\&D/assets$, $Patent\ index$, $\Delta R\&D/assets$, and $\Delta Patent\ index$. These capture both levels and trends in firms' innovation activities.

Second, I incorporate the insight of Phillips and Zhdanov (2013), who argue that large, mature firms tend to outsource R&D through acquisitions. To reflect this, I include measures of acquirer size and maturity: *Log asset*, *Patent stock*, and firm *Age*. These variables also indirectly speak to the view that managers of entrenched firms may use acquisitions defensively to preempt future takeovers, as argued by Gorton, Kahl, and Rosen (2009).

Third, I consider neoclassical acquisition motives. According to Jovanovic and Rousseau (2002) and Maksimovic and Phillips (2001), firms with higher productivity tend to acquire less productive ones to allocate capital more efficiently. I measure productivity via $Tobin's\ Q$ and a standard estimate of total factor productivity (TFP), assuming a Cobb-Douglas production function with capital and labor inputs. Notably, $Tobin's\ Q$ also reflects valuation-driven motives, as emphasized in Dong et al. (2006), Rhodes-Kropf, Robinson, and Viswanathan (2005), and Shleifer and Vishny (2003).

Fourth, I account for agency-based motives. Jensen (1986) theorizes that managers may undertake acquisitions to dissipate excess free cash flows rather than return them to shareholders. Harford (1999) provide empirical support for this claim. To capture this perspective,

I include measures of financial slack: $Cashflow/asset$, $Cash/asset$, and $Market\ leverage$. These variables also reflect the enabling role of liquidity in driving acquisition waves, as documented by Harford (2005).

Fifth, I consider internal investment opportunities as a motive. According to Levine (2017), firms may pursue acquisitions when they lack investment opportunities internally. I include $Capex/asset$ and $Sales\ growth$ to capture this logic, reflecting the firm's investment intensity and growth perspective.

Sixth, competition may pressure firms to acquire external innovation or suppress rivals. Higgins and Rodriguez (2006) document that pharmaceutical firms use acquisitions to renew their innovation pipelines in the face of patent expirations. Cunningham, Ederer, and Ma (2021) find that firms engage in "killer acquisitions" to preempt emerging competitors. To reflect industry conditions, I use two Herfindahl indices: one based on product-market sales ($Competition$), and the other based on patent stock ($Tech\ competition$). Both are transformed as one minus the standard Herfindahl index, so that higher values reflect greater competition.

Lastly, I incorporate measures of managerial overconfidence, in line with the hubris hypothesis of Roll (1986) and further developed by Malmendier, Opp, and Saidi (2016). Following Barber and Odean (2001) and Huang and Kisgen (2013), I use the presence of a female executive ($Female\ executive$) or a female CEO ($Female\ CEO$) as inverse proxies for overconfidence based on robust evidence that male managers are, on average, more overconfident in financial decision-making.

4.3.2 Decomposing the gap measure

To investigate whether the effect of *GapSignal* is truly driven by informational surprises, I examine to what extent it can be explained by firm characteristics previously identified as acquisition antecedents. To that end, I estimate the following regression model:

$$GapSignal_{id} = \mathbf{\Theta}_{id} \cdot \mathbf{\Xi} + \alpha_u + \alpha_v + \alpha_t + \varepsilon_{id}, \tag{31}$$

where i indexes acquirers, d indexes acquisition deals, u and v denote 2-digit SIC industry of the acquirer and the target, respectively, and t represents the calendar year. $\mathbf{Z}_{id}$ is a vector of acquisition antecedents, as listed in Section 4.3.1. The regression includes industry and year fixed effects α_u, α_v, α_t to account for unobserved heterogeneity across industries and time.

I estimate the model using the sample of public-target acquisitions, since private targets, on average, reveal little information about the acquirer's technological position. Table 6

presents the results.

Column (1) shows that higher R&D intensity is strongly associated with higher *GapSignal* scores, while increases in R&D intensity predict lower scores, suggesting that gap bidders' acquisition motives are distinct from those documented by Bena and Li (2014). Rather, gap bidders tend to be R&D-intensive companies facing a decline in R&D investment—probably due to a lack of internal growth opportunities. Column (2) indicates that larger and younger acquirers have higher *GapSignal* scores, only partially consistent with theories of strategic outsourcing. Column (3) demonstrates that more productive acquirers are more frequently gap bidders, suggesting that gap bidders' acquisition motives overlap with those predicted by neoclassical acquisition theories. In Column (4), market leverage is negatively associated with *GapSignal*, suggesting financial constraints dampen gap-closing acquisitions.

Column (6) shows that acquirers in more competitive product markets are less likely to become gap bidders, contrary to strategic pressure arguments, but more in line with a Schumpetarian growth theory where product market competition erodes the monopolistic rents accruing to successful innovators, hence making cutting-edge technology less attractive (Aghion & Howitt, 1992). Column (7) finds no significant effect of managerial gender traits. Column (8) further rules out cash flow-driven agency motives and indicates that firms with weaker growth prospects are more inclined to close technological gaps via acquisition. Overall, while observable predictors explain some variation in *GapSignal*, a substantial portion remains unexplained.

Equation (31) enables a decomposition of *GapSignal* into two parts: a fitted component explained by known acquisition predictors, and a residual component orthogonal to them. Table 6 replaces *GapSignal* with these two components and re-estimates Equation (28). The results reveal that only the residual component significantly predicts acquirer CARs, suggesting that market reactions are primarily driven by the unanticipated part of the technological gap—i.e., information not already embedded in known firm and industry fundamentals.

4.4 Market reactions for industry peers

In addition to acquirer returns, the model predicts a positive relationship between the market reactions of the acquirer's peers and *GapSignal* (Hypothesis 3 in Section 2.3.4). This section tests this hypothesis.

I begin by constructing a sample of peer firms for the acquirers from the universe of COMPUSTAT firms. I exclude firms with no patent throughout the sample period and, for each acquirer, extract all firm-years within the same two-digit SIC industry in the announcement year. I then exclude firms that announce any acquisitions during years $t-3$ to t, where t denotes the year of the focal announcement. Finally, I compute ARs for peer firms over the

same event windows used for acquirers.

I then regress peers' announcement CARs on the corresponding acquisition's *GapSignal*, controlling for deal characteristics and peer fundamentals (including *Log size*, *Log Tobin's Q*, *Market leverage* and *Log patent stock*). I include industry fixed effects based on peers' two-digit SIC codes and year fixed effects, or alternatively industry–year fixed effects, to account for unobserved heterogeneity. Standard errors are clustered at the deal level. Panel A of Table 8 reports the results.

Columns 1–4 all yield positive coefficients, consistent with the prediction that a larger revealed gap, and hence a more negative revaluation of the acquirer, is accompanied by a more positive revaluation of its industry peers. The coefficients are also statistically significant for the narrower event window ($CAR(-1,1)$ in columns 1–2), implying that a one-standard-deviation increase in *GapSignal* is associated with a 15 bp increase in peers' announcement returns. The estimates are substantially weaker when the dependent variable is $CAR(-3,3)$, possibly because the wider event window introduces additional noise that attenuates the effect.

To rule out the possibility that this positive correlation might be driven by expectations of future involvement in acquisitions, I construct two indicators, *Target in* 3 *years* and *Acquirer in* 3 *years*, which equal one if the peer firm becomes a target or an acquirer within three years subsequent to the announcement, and zero otherwise. I then taken them as the dependent variable. Panel B reports the results. The small point estimates and weak significance across the board does not support this interpretation.

To rule out the possibility that this positive correlation is driven by expectations of future acquisition involvement, I construct two indicators, *Target in* 3 *years* and *Acquirer in* 3 *years*, which equal one if a peer firm becomes a target or an acquirer within three years following the announcement, and zero otherwise. I then use these indicators as dependent variables. Panel B reports the results. The small point estimates and uniformly weak statistical significance do not support this interpretation.

Taken together, these results are consistent with Hypothesis 3, suggesting that, in environments characterized by informational frictions, acquisition announcements have value implications not only for the transacting parties but also for the broader industry, through shifts in the market's prior beliefs about the distribution of technological capabilities.

5 Extensions and robustness

In this section, I show that the results persist for at least one year after the announcement, rule out alternative explanations such as dissynergies, overpayment, and anticipation,

demonstrate the robustness of the main findings to alternative definitions of the revealed technological gap, and examine whether acquisition announcements also reveal information about the targets.

5.1 Long-term effect

To examine whether the negative correlation between *GapSignal* and short-window market reactions persists over longer horizons, I calculate acquirer announcement CARs over $h \in \{10, 20, \ldots, 250\}$ trading days following the announcements, and estimate the following regression model for each horizon:

$$CAR(-3, h)_{id} = \gamma \cdot GapSignal_d + \mathbf{X}_{id} \cdot \mathbf{\Gamma} + \alpha_u + \alpha_v + \alpha_t + \varepsilon_{id}, \tag{32}$$

where i indexes acquirers, d indexes acquisition deals, u and v denote 2-digit SIC industry of the acquirer and the target, respectively, and t represents the calendar year. $\mathbf{X}_{id}$ is a vector of control variables introduced in Section 3.2.3.

The estimated coefficients over different horizons are plotted in Figure 7. Across all horizons, the point estimates remain persistently negative, and their magnitude increases with the horizon, from -0.06 at $h = 10$ to -0.17 at $h = 250$. This pattern is consistent with a drift, likely reflecting the market's gradual learning about the target's technological profile. The coefficients are statistically significant at the 5% level up to 160 trading days after the announcement and remain significant at the 10% level through $h = 250$.

I also stack all horizons into a single regression, additionally including horizon fixed effects and clustering standard errors at the transaction level, since overlapping horizons induce strong within-transaction correlations. The estimated coefficient is -0.13 and significant at the 5% level. Therefore, during the year after acquisition announcements, a one standard deviation increase in *GapSignal* is associated with a 4% drop in acquirers' market value.

Therefore, the revaluation effect documented in the preceding sections remains persistent and economically meaningful for at least one year after the announcement.

5.2 Alternative explanations

This section considers three alternative explanations for the observed negative correlation between acquirer returns and *GapSignal*, which proxies for the acquirer's revealed technological gap:

(a) dissynergies due to technological incompatibility between the acquirer and the target.

(b) overpayment for technologically advanced targets; and

(c) pre-announcement anticipation, where expected gains are partially priced in, resulting in seemingly more negative reactions at the time of the announcement.

5.2.1 Dissynergies

The measure *GapSignal* captures the similarity between the target's technological portfolio and that of the acquirer's industry peers. A natural alternative interpretation, then, is that high *GapSignal* reflects an acquirer's tendency to imitate the technological direction of its peers—potentially to its own detriment. In this view, the measure may capture strategic misalignment or inefficiency arising from blind conformity, where the acquirer pursues targets without critically assessing whether the target is compatible with its own capabilities or long-term strategy. Under this hypothesis, the negative correlation between acquirer CARs and *GapSignal* reflects growing dissynergies arising from a widening mismatch between the acquirer and the target.

To address this concern, I examine withdrawn deals, in which any (dis)synergies expected to accrue to the acquirer should reverse following withdrawal. If dissynergies were driving my findings, the negative correlation between acquirer returns and *GapSignal* should disappear after withdrawal, or at least weaken if the market still anticipates that the acquirer may pursue similar acquisitions in the future.

By contrast, if revaluations were driving my findings, I would expect withdrawals to strengthen the negative correlation between acquirer returns and *GapSignal*. The model implies that the documented correlation reflects the net effect of two forces: revaluations due to the revealed technological gap and unexpected synergies that would otherwise be unattainable without the gap (see Section A.1.3 in the Appendix). While the latter partially offsets the negative valuation impact, the former dominates. Withdrawals reverse the synergy component but leave the revaluation component intact, since revealed information cannot be undone. Consequently, withdrawals should amplify the effect.

Figure 8 shows the evolution of sample deals up to 250 trading days after the announcement. By 70, 120, and 220 trading days post-announcement, 50%, 80%, and 90% of deals, respectively, have been either consummated or terminated. By the end of the period, 77% of deals are effective and 16% are withdrawn.

Accordingly, I split the long-term CARs into three subsamples: effective, pending, and withdrawn. Within each subsample, I stack different horizons and estimate Equation (32), including horizon fixed effects and clustering standard errors at the transaction level. Table 9 reports the results. The effective and pending subsamples yield coefficients on *GapSignal* of similar magnitude, although the coefficient for the effective group is not statistically significant. By contrast, the coefficient for the withdrawn subsample is both statistically significant

and substantially larger in magnitude—nearly five times that of the other two groups. These results strongly support revaluations, rather than dissynergies, as the primary driver of my findings.

5.2.2 Overpayment

A desperate gap bidder may overpay for a target that helps close its technological gap—especially when the gap is large and the urgency to catch up is high. This puts acquirers with larger gaps in a weaker bargaining position relative to those facing smaller gaps. If this mechanism is at play, overpayment may be misinterpreted as negative revaluation, potentially confounding the main interpretation of the market reaction.

To address this possibility, I examine the relationship between target announcement returns and *GapSignal*. The rationale is straightforward: if acquirers systematically overpay to close their technological gaps, this overpayment would constitute a wealth transfer from acquirer shareholders to target shareholders. Consequently, if overpayment were driving the results, the observed negative correlation between acquirer returns and *GapSignal* should be mirrored by a *positive* correlation between target returns and *GapSignal*.

Table 10 reports the results. Across all specifications and both event windows, the coefficient on *GapSignal* is statistically insignificant and consistently close to zero. If anything, the point estimates are negative, opposite to what the overpayment hypothesis would predict. This finding suggests that the acquirer's revealed technological gap does not predict target returns, providing no evidence that acquirers systematically overpay in gap-closing acquisitions.

Control variables. Several controls merit attention. First, *Alignment*—which captures the acquirer's technological overlap with its industry—is negatively associated with target CARs, indicating that better-positioned acquirers may enjoy greater bargaining power. Second, the indicator for tender offers is positively and significantly associated with target returns, consistent with prior findings that unsolicited bids often carry higher acquisition premiums (Bradley, Desai, & Kim, 1988; Walkling, 1985). Finally, target size (*Log asset (T)*) is positively related to announcement returns, possibly reflecting the stronger negotiation leverage of larger targets.

5.2.3 Anticipation

Prior research shows that market anticipation can partially incorporate expected acquisition gains into stock prices before deal announcements, thereby dampening the observed acquirer returns at the time of the announcement (Cai, Song, & Walkling, 2011; Tunyi, 2021). If

GapSignal also correlates with the extent to which the market expects a firm to pursue a gap-closing acquisition, then acquirers with higher *GapSignal* values may exhibit lower announcement returns—not due to the revelation of new information, but because a greater portion of the expected gains has already been priced in.

To address this alternative explanation, I argue that pre-announcement anticipation tends to reduce the element of surprise at acquisition announcements. The greater the anticipation, the smaller the informational shock, and consequently, the more muted the market reaction. Therefore, if *GapSignal* primarily captures market anticipation rather than new information, a higher *GapSignal* should be associated with a smaller absolute magnitude of CARs, regardless of direction.

In this light, I regress the absolute value of acquirer CARs against *GapSignal*. Table 11 presents the results. Panel A reports the results for the event window CAR(-1,1), and Panel B for CAR(-3,3). Columns (1) and (2) reveal a positive and statistically significant relationship between *GapSignal* and |CAR| If *GapSignal* merely captured pre-announcement anticipation, I would expect to see smaller absolute returns as *GapSignal* increases, reflecting muted surprise. Instead, the results suggest that larger revealed technological gaps lead to stronger market reactions, which is inconsistent with the prediction of an anticipation-based explanation.

Columns (3) through (6) split the sample based on whether CARs are negative or non-negative. The negative relationship between *GapSignal* and acquirer returns is concentrated entirely among deals with negative acquirer CARs—those that likely triggered the strongest negative revaluation. In contrast, *GapSignal* has no meaningful effect when CARs are non-negative, suggesting that revaluation is largely absent in those cases.

In summary, the negative relationship between *GapSignal* and acquirer returns is unlikely driven by market anticipation.

5.3 Redefining the gap measure

More granular patent classification The earlier analyses define *GapSignal*—the technological overlap between the target and the acquirer's industry—using 137 technological classes derived from the *class* level of the CPC classification system. To ensure that the main results are not sensitive to the granularity of the classification scheme, I re-estimate *GapSignal* based on the more detailed *subclass* level, which includes 680 categories.

With the redefined *GapSignal*—constructed using CPC subclass codes—I replicate the baseline analysis from Table 2. Table 12, Panel A presents the results. The estimates are both qualitatively and quantitatively similar to the baseline, confirming that the negative relationship between *GapSignal* and acquirer announcement returns is robust to alternative

definitions of the revealed technological gap. In both event windows—CAR(-1,1) and CAR(-3,3)—the coefficients on *GapSignal* remain negative and statistically significant across all specifications. However, the results in Panel A show somewhat weaker statistical significance in Columns (1) to (3). This attenuation likely arises from the assumption implicit in cosine similarity that all dimensions are mutually orthogonal, even though the technological distance between patent categories may vary. As the classification scheme becomes more granular, this assumption introduces greater measurement noise, potentially weakening the informativeness of the signal extracted from the target's patent portfolio. Nonetheless, the overall findings remain consistent and robust.

Shorter patenting histories. In the earlier analyses, *GapSignal* is defined using the 10-year patenting histories of the target firms and the acquirer's collective industry. While using a longer history provides more observations, thereby mitigating measurement error, it may also bias the measure toward outdated technologies. To address this concern, I redefine *GapSignal* using 5-year patenting histories.

Table 12, Panel B, reports the results, replicating Table 2. Across all specifications, the coefficient on this version of *GapSignal* is negative, consistent with the baseline results, albeit with slightly weaker statistical significance. This pattern is likely attributable to increased measurement error from estimating the measure using only half of the patent observations. Overall, the choice of the patent-history window is unlikely to materially drive the documented effect.

Patent quality. In the earlier analyses, technological profiles are characterized by the frequency of patenting activity across technological fields. This choice is motivated by the argument that differential activity across categories can reflect technological trends. However, it also implicitly treats each patent as carrying the same economic value, which is unrealistic. In other words, *GapSignal* captures patent quantity while abstracting entirely from patent quality.

The baseline analysis does not incorporate quality because satisfactory measures are difficult to obtain. This section therefore leverages two imperfect proxies for patent quality to address this concern. The first measure is the number of citations each patent receives up to 2022. Two issues arise with this measure. First, citations are right-truncated: older patents accumulate more citations, not necessarily because they are of higher quality, but because they have had more time to be cited. I follow Hall, Jaffe, and Trajtenberg (2001) in addressing this issue by dividing each patent's citation count by the mean citation count of its application-year cohort. Second, citations may introduce look-ahead bias by incorporating

ex post information. I attempt to address this second concern using an alternative measure.

I recompute the gap measure by weighting each patent by its cohort-adjusted citation count. Table 12, Panel C, reports the results. Across all specifications, the coefficient on the newly defined *GapSignal* is negative and statistically significant, consistent with the baseline.

The second quality proxy I use is the real private economic value of patents to firms, as developed by Kogan et al. (2017). This measure leverages stock market responses to patent news and therefore does not rely on ex post information. Unfortunately, by construction it only covers patents filed by public firms. Using this measure therefore restricts the patent portfolio collectively held by the acquirer's industry to those of public firms.

I then recompute the gap measure by weighting each patent by the logarithm of its real economic value. Table 12, Panel D, reports the results. The coefficient on the newly defined *GapSignal* is negative across all specifications, although the statistical significance is weaker than in the baseline. This pattern may be attributable to the exclusion of private firms from the measure.

Taken together, the main findings of the paper are robust to all four alternative definitions of the gap measure.

5.4 Are targets revalued?

Thus far, this paper has demonstrated that an acquirer's choice of target can reveal the acquirer's weaknesses and trigger negative revaluation. A natural question that follows is whether the reverse inference is also possible: can the market infer hidden characteristics of the target firm based on observable features of the acquirer?

The M&A market is increasingly understood through the lens of two-sided matching frameworks.[17] In such models, the formation of acquirer-target pairs reflects the preferences of both parties, suggesting that observed matches embed information about the underlying—potentially hidden—characteristics that shape these preferences. Supporting this perspective, Wang (2018) find that revaluation effects explain approximately 26% of the variation in target announcement returns. Notably, however, the share is substantially higher on the acquirer side, where revaluation accounts for 58% of the return variation. This asymmetry suggests that, while target characteristics matter, observed deal outcomes are more strongly shaped by the acquirer's preferences and informational content, likely due to their larger size and greater bargaining power.

To empirically examine whether partnering with an acquirer that helps the target close its technological gap leads to negative revaluation of the target, I construct target-side analogs of

[17] See, for example, Akkus, Cookson, and Hortaçsu (2016) and Fox (2018).

the *GapSignal* and *Alignment* measures. Specifically, *GapSignal (T)* captures the alignment between the acquirer's technological profile and the target's industry technology set, reflecting the extent to which the acquirer helps fill the target's technological gap. *Alignment (T)*, by contrast, measures the similarity between the target's own technological portfolio and the technology set of its industry, capturing the target's standalone technological positioning.

Table 13 reports the results. In the baseline models—Columns (1) and (4)—target *GapSignal* is negatively associated with target CARs, with significance at the 5% and 10% levels, respectively. However, once more stringent fixed effects are introduced, the statistical significance dissipates, suggesting that the effect is not particularly robust. The other tabulated variables are uniformly insignificant across specifications. Taken together, the results provide only suggestive evidence that targets are revalued downward when the deal implies that the acquirer is filling a technological gap for the target. However, this effect—if present—is considerably weaker and less systematic than the revaluation observed on the acquirer side.

6 Conclusion

This paper reframes how we interpret market reactions to acquisition announcements by emphasizing that deals can systematically reveal hidden weaknesses about the acquirer. Acquirers with disadvantages—such as technological deficiencies examined in this study—may self-select into acquisitions as a means of closing competitive gaps, making the act of acquisition itself informative about the acquirer's disadvantageous competitive standing. Specifically, in technology-oriented acquisitions, a target's technological profile serves as a signal: when the target greatly overlaps with the acquirer's industry technologically, the market infers that the acquirer is lagging behind, triggering negative revaluation.

The strength of this revaluation, however, depends on the degree of informational frictions. When the target is more opaque than the acquirer—as is typically the case with private targets—the market cannot extract reliable inferences based on the target's observable traits, and the revaluation effect is muted. This asymmetry offers a novel explanation for the longstanding puzzle that acquirers of private targets tend to earn positive announcement returns: not because the deals are better, but because they reveal less negative information. More broadly, these findings underscore the importance of informational dynamics—alongside deal fundamentals—in shaping M&A outcomes.

References

Abdi, F., & Ranaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *The Review of Financial Studies*, *30*(12), 4437–4480.

Aghion, P., & Howitt, P. (1992). A model of growth through creative destruction. *Econometrica*, *60*(2), 323–351.

Akbulut, M. E., & Matsusaka, J. G. (2010). 50+ years of diversification announcements. *Financial Review*, *45*(2), 231–262.

Akhigbe, A., Martin, A. D., & Whyte, A. M. (2007). Partial acquisitions, the acquisition probability hypothesis, and the abnormal returns to partial targets. *Journal of Banking & Finance*, *31*(10), 3080–3101.

Akkus, O., Cookson, J. A., & Hortaçsu, A. (2016). The determinants of bank mergers: A revealed preference analysis. *Management Science*, *62*(8), 2241–2258.

Aktas, N., de Bodt, E., & Roll, R. (2011). Serial acquirer bidding: An empirical test of the learning hypothesis. *Journal of Corporate Finance*, *17*(1), 18–32.

Anderson, R. C., Duru, A., & Reeb, D. M. (2009). Founders, heirs, and corporate opacity in the united states. *Journal of Financial Economics*, *92*(2), 205–222.

Ang, J., & Kohers, N. (2001). The take-over market for privately held companies: The US experience. *Cambridge Journal of Economics*, *25*(6), 723–748.

Ang, J. S., & Cheng, Y. (2006). Direct evidence on the market-driven acquisition theory. *Journal of Financial Research*, *29*(2), 199–216.

Barber, B. M., & Odean, T. (2001). Boys will be boys: Gender, overconfidence, and common stock investment*. *The Quarterly Journal of Economics*, *116*(1), 261–292.

Bena, J., & Li, K. (2014). Corporate innovations and mergers and acquisitions. *The Journal of Finance*, *69*(5), 1923–1960.

Ben-David, I., Bhattacharya, U., Huang, R., & Jacobsen, S. E. (2025, April 30). The (missing) relation between acquisition announcement returns and value creation.

Bessen, J. (2003). Patent thickets: Strategic patenting of complex technologies. *Faculty Scholarship*.

Betton, S., Eckbo, B. E., & Thorburn, K. S. (2008). CORPORATE TAKEOVERS.

Blouin, J. L., Fich, E. M., & Tran, A. L. (2020). Documenting m&a's revelation effect using state-level r&d tax incentives.

Boehm, B. W. (1981). *Software engineering economics*. Prentice-Hall.

Bradley, M., Desai, A., & Kim, E. H. (1988). Synergistic gains from corporate acquisitions and their division between the stockholders of target and acquiring firms. *Journal of Financial Economics*, *21*(1), 3–40.

Brooks, J., Frederick P. (1975). *The mythical man-month: Essays on software engineering*. Addison-Wesley.

Cai, J., Song, M. H., & Walkling, R. A. (2011). Anticipation, acquisitions, and bidder returns: Industry shocks and the transfer of information across rivals. *The Review of Financial Studies*, *24*(7), 2242–2285.

Cai, X., De Cesari, A., Gao, N., & Peng, N. (2024). Acquisitions and technology value revision. *Management Science*, *70*(7), 4283–4305.

Cappelli, R., Corsino, M., Laursen, K., & Torrisi, S. (2023). Technological competition and patent strategy: Protecting innovation, preempting rivals and defending the freedom to operate. *Research Policy*, *52*(6), 104785.

Capron, L., & Shen, J.-C. (2007). Acquisitions of private vs. public firms: Private information, target selection, and acquirer returns. *Strategic Management Journal*, *28*(9), 891–911.

Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, *52*(1), 57–82.

Chang, S. (1998). Takeovers of privately held targets, methods of payment, and bidder returns. *The Journal of Finance*, *53*(2), 773–784.

Cheng, E. C. M., Courtenay, S. M., & Krishnamurti, C. (2006). The impact of increased voluntary disclosure on market information asymmetry, informed and uninformed trading. *Journal of Contemporary Accounting & Economics*, *2*(1), 33–72.

Cunningham, C., Ederer, F., & Ma, S. (2021). Killer acquisitions. *Journal of Political Economy*, *129*(3), 649–702.

Daske, H., Hail, L., Leuz, C., & Verdi, R. (2008). Mandatory IFRS reporting around the world: Early evidence on the economic consequences. *Journal of Accounting Research*, *46*(5), 1085–1142.

Derrien, F., Frésard, L., Slabik, V., & Valta, P. (2023). Industry asset revaluations around public and private acquisitions. *Journal of Financial Economics*, *147*(1), 243–269.

Dong, M., Hirshleifer, D., Richardson, S., & Teoh, S. H. (2006). Does investor misvaluation drive the takeover market? *The Journal of Finance*, *61*(2), 725–762.

Faccio, M., McConnell, J. J., & Stolin, D. (2006). Returns to acquirers of listed and unlisted targets. *Journal of Financial and Quantitative Analysis*, *41*(1), 197–220.

Fama, E. F. (1998). Market efficiency, long-term returns, and behavioral finance1. *Journal of Financial Economics*, *49*(3), 283–306.

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56.

Farre-Mensa, J., Hegde, D., & Ljungqvist, A. (2020). What is a patent worth? evidence from the u.s. patent "lottery". *The Journal of Finance*, *75*(2), 639–682.

Fox, J. T. (2018). Estimating matching games with transfers. *Quantitative Economics*, *9*(1), 1–38.

Franks, J., Harris, R., & Titman, S. (1991). The postmerger share-price performance of acquiring firms. *Journal of Financial Economics*, *29*(1), 81–96.

Fuller, K., Netter, J., & Stegemoller, M. (2002). What do returns to acquiring firms tell us? evidence from firms that make many acquisitions. *The Journal of Finance*, *57*(4), 1763–1793.

Gorton, G., Kahl, M., & Rosen, R. J. (2009). Eat or be eaten: A theory of mergers and firm size. *The Journal of Finance*, *64*(3), 1291–1344.

Gurgula, O. (2020). Strategic patenting by pharmaceutical companies – should competition law intervene? *Iic; International Review of Industrial Property and Copyright Law*, *51*(9), 1062–1085.

Hall, B., Jaffe, A., & Trajtenberg, M. (2001). The NBER patent citations data file: Lessons, insights and methodological tools. *Tel Aviv, Papers*, *8498*.

Hall, B. H., Thoma, G., & Torrisi, S. (2007). The market value of patents and r&d: Evidence from european firms. *Academy of Management Proceedings*, *2007*(1), 1–6.

Hansen, R. G., & Lott, J. R. (1996). Externalities and corporate objectives in a world with diversified shareholder/consumers. *Journal of Financial and Quantitative Analysis*, *31*(1), 43–68.

Harford, J. (1999). Corporate cash reserves and acquisitions. *The Journal of Finance*, *54*(6), 1969–1997.

Harford, J. (2005). What drives merger waves? *Journal of Financial Economics*, *77*(3), 529–560.

Harrell, F. E. (2001). *Regression modeling strategies: With applications to linear models, logistic regression, and survival analysis*. Springer New York.

Hegde, D., Herkenhoff, K., & Zhu, C. (2023). Patent publication and innovation. *Journal of Political Economy*, *131*(7), 1845–1903.

Hietala, P., Kaplan, S. N., & Robinson, D. T. (2002, October). What is the price of hubris? using takeover battles to infer overpayments and synergies.

Higgins, M. J., & Rodriguez, D. (2006). The outsourcing of r&d through acquisitions in the pharmaceutical industry. *Journal of Financial Economics*, *80*(2), 351–383.

Hoberg, G., & Phillips, G. (2010). Product market synergies and competition in mergers and acquisitions: A text-based analysis. *The Review of Financial Studies*, *23*(10), 3773–3811.

Hong, H., Lim, T., & Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, *55*(1), 265–295.

Huang, J., & Kisgen, D. J. (2013). Gender and corporate finance: Are male executives overconfident relative to female executives? *Journal of Financial Economics*, *108*(3), 822–839.

Huberman, B. A., & Wilkinson, D. M. (2005). Performance variability and project dynamics. *Computational Economics*, *25*(1-2), 3–16.

Jacobsen, S. (2014). The death of the deal: Are withdrawn acquisition deals informative of CEO quality? *Journal of Financial Economics*, *114*(1), 54–83.

Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. *The American Economic Review*, *76*(2), 323–329.

Jia, X., & Sun, S. T. (2022, November 25). Demand complementarity and mergers and acquisitions.

Jovanovic, B., & Rousseau, P. L. (2002). The q-theory of mergers. *The American Economic Review*, *92*(2), 198–204.

Kogan, L., Papanikolaou, D., Seru, A., & Stoffman, N. (2017). Technological innovation, resource allocation, and growth*. *The Quarterly Journal of Economics*, *132*(2), 665–712.

Lee, K. H., Mauer, D. C., & Xu, E. Q. (2018). Human capital relatedness and mergers and acquisitions. *Journal of Financial Economics*, *129*(1), 111–135.

Leuz, C., & Verrecchia, R. E. (2000). The economic consequences of increased disclosure. *Journal of Accounting Research*, *38*, 91–124.

Levine, O. (2017). Acquiring growth. *Journal of Financial Economics*, *126*(2), 300–319.

Loderer, C., & Martin, K. (1990). Corporate acquisitions by listed firms: The experience of a comprehensive sample. *Financial Management*, *19*(4), 17–33.

Maksimovic, V., & Phillips, G. (2001). The market for corporate assets: Who engages in mergers and asset sales and are there efficiency gains? *The Journal of Finance*, *56*(6), 2019–2065.

Malmendier, U., Opp, M. M., & Saidi, F. (2016). Target revaluation after failed takeover attempts: Cash versus stock. *Journal of Financial Economics*, *119*(1), 92–106.

Mitchell, M. L., & Stafford, E. (2000). Managerial decisions and long-term stock price performance. *The Journal of Business*, *73*(3), 287–329.

Moeller, S. B., Schlingemann, F. P., & Stulz, R. M. (2004). Firm size and the gains from acquisitions. *Journal of Financial Economics*, *73*(2), 201–228.

Moeller, S. B., Schlingemann, F. P., & Stulz, R. M. (2005). Wealth destruction on a massive scale? a study of acquiring-firm returns in the recent merger wave. *The Journal of Finance*, *60*(2), 757–782.

Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, *13*(2), 187–221.

Pakes, A. (1985). On patents, r & d, and the stock market rate of return. *Journal of Political Economy*, *93*(2), 390–409.

Phillips, G. M., & Zhdanov, A. (2013). R&d and the incentives from merger and acquisition activity. *The Review of Financial Studies*, *26*(1), 34–78.

Reeb, D. M., & Zhao, W. (2020, February 18). Patents do not measure innovation success.

Rhodes-Kropf, M., & Robinson, D. T. (2008). The market for mergers and the boundaries of the firm. *The Journal of Finance*, *63*(3), 1169–1211.

Rhodes-Kropf, M., Robinson, D. T., & Viswanathan, S. (2005). Valuation waves and merger activity: The empirical evidence. *Journal of Financial Economics*, *77*(3), 561–603.

Roll, R. (1986). The hubris hypothesis of corporate takeovers. *The Journal of Business*, *59*(2), 197–216.

Savor, P. G., & Lu, Q. (2009). Do stock mergers create value for acquirers? *The Journal of Finance*, *64*(3), 1061–1097.

Seru, A. (2014). Firm boundaries matter: Evidence from conglomerates and r&d activity. *Journal of Financial Economics*, *111*(2), 381–405.

Shleifer, A., & Vishny, R. W. (2003). Stock market driven acquisitions. *Journal of Financial Economics*, *70*(3), 295–311.

Travlos, N. G. (1987). Corporate takeover bids, methods of payment, and bidding firms' stock returns. *The Journal of Finance*, *42*(4), 943–963.

Tunyi, A. A. (2021). Revisiting acquirer returns: Evidence from unanticipated deals. *Journal of Corporate Finance*, *66*, 101789.

Walkling, R. A. (1985). Predicting tender offer success: A logistic analysis. *Journal of Financial and Quantitative Analysis*, *20*(4), 461–478.

Wang, W. (2018). Bid anticipation, information revelation, and merger gains. *Journal of Financial Economics*, *128*(2), 320–343.

A Appendix

A.1 Derivations

A.1.1 Important thresholds in the optimal investment strategy

The model shows that gap bidders and non-gap bidders can be defined based on their initial technology endowment z. Specifically, when $z \in [z^*, z^{max}]$, the firm buys technology units externally and becomes a gap bidder. When $z \in [0, z^*)$, the firm builds all technology unit internally. Accordingly, gap bidders' optimal gross investment ranges in $I \in (I^*, I^{max}]$, while that of non-gap bidders' ranges in $I \in [0, I^*]$ This appendix defines these threshold values.

First, the critical endowment level z^* distinguishes gap bidders from non-gap bidders. It corresponds to the critical gross investment I^*. It is given by

$$z^* = \left(\frac{\alpha\kappa}{p+f+\beta}\right)^{\frac{1}{1-\alpha}} - I^*. \tag{33}$$

Second, since the optimal gross investment strictly decreases with z, the maximum gross investment $I^{\max}$ can be obtained at zero endowment ($z = 0$):

$$I^{max} = \left(\frac{\alpha\kappa}{p+f+\beta}\right)^{\frac{1}{1-\alpha}}. \tag{34}$$

Third, I define the maximum level of endowment $z^{\max}$ as the level corresponding to zero gross investment. It provides a natural upper bound for z:

$$z^{max} = \left(\frac{\alpha\kappa}{p}\right)^{\frac{1}{1-\alpha}}. \tag{35}$$

A.1.2 The optimal investment strategy

The optimal investment strategy is obtained when the built technology units x and the bought units y maximizes firm value V, given the technology endowment z. Expressing x

and y as functions of z yields

$$x(z) = \begin{cases} I^* & \text{if } z \in [0, z^*), \\ x^s(z) & \text{if } z \in [z^*, z^{max}], \end{cases} \tag{36}$$

$$y(z) = \begin{cases} I^{max} - I^* - z & \text{if } z \in [0, z^*), \\ 0 & \text{if } z \in [z^*, z^{max}], \end{cases} \tag{37}$$

$$\text{where} \quad x^s(z) = \arg\max_{x \geq 0} Q(z + x) - c(x, 0). \tag{38}$$

The function $c(\cdot, \cdot)$ specifies the units built and units bought separately. The function $x^s(z)$ describes non-gap bidders' optimal number of units built. Effectively, it is identical to gap bidders' optimal number of units built when buying is unavailable (i.e., subject to the constraint $y = 0$), since the constraint always binds for non-gap bidders.

Given the optimal strategy, firm value can be written as a function of z:

$$V(z) = Q(z + x(z) + y(z)) - c(x(z), y(z)). \tag{39}$$

Firm value when buying is unavailable is

$$V^s(z) = Q(z + x^s(z)) - c(x^s(z), 0). \tag{40}$$

Notably, for gap bidders, $V^s(z)$ is steeper than $V(z)$—i.e., $V^{s\prime}(z) > V'(z)$. When $z \in [0, z^*)$, it can be shown that

$$V(z) = Q(I^{max}) - (p + f + \beta)(I^{max} - I^* - z),$$

which implies $V'(z) = p + f + \beta$. Similarly, for $z \in [0, z^*)$,

$$V^s(z) = Q(z + x^s(z)) - c(x^s(z), 0),$$

so that $V^{s\prime}(z) = Q'(z + x^s(z))$.

Since the constraint $y = 0$ is binding for gap bidders, $x^s(z) + z < x(z) + y(z) + z = I^{max}$. Intuitively, when acquisition is unavailable, gap bidders must substitute relatively cheaper purchased technology with more expensive in-house investment, leading to a lower final technology level. Under decreasing returns to scale, it follows that

$$Q'(I^{max}) = p + f + \beta < Q'(z + x^s(z))$$

Therefore, $V^{s\prime}(z) > V'(z)$ for $z \in [0, z^*)$.

A.1.3 Decomposing announcement abnormal returns

Gap bidders' announcement AR consist of two components: revaluations due to the revelation of their weak technological standing, and unexpected acquisition synergies. I now decompose it into these two components.

Revaluations. Revaluations capture changes in the market's assessment of a gap bidder's *standalone* value, isolating the informational impact of the acquisition while abstracting from its realized economic outcomes. The key is to construct the counterfactual market value that would obtain if the market incorporated the information revealed by the acquisition in the absence of the transaction itself. This can be implemented by assuming all acquisitions are reversed after the announcement, which effectively imposes $y = 0$.

Accordingly, a gap bidder's pre-announcement standalone value can be expressed as

$$V_0^s(z) = p_A \cdot V^s(z) + (1 - p_A) \cdot \overline{V}^s, \tag{41}$$

where V^s denotes a firm's intrinsic value under the optimal investment strategy when acquisition is unavailable. $\overline{V}^s$ stands for the average intrinsic value across all firms in this scenario.

In the presence of the information revealed through the acquisition announcement, a gap bidder's post-announcement standalone value can be expressed as

$$V_1^s(z) = p_{AT} \cdot V^s(z) + (1 - p_{AT}) \cdot \left[p_T \cdot \overline{V}_G^s \cdot (1 - p_T) + \overline{V}^s \right], \tag{42}$$

where $\overline{V}_G^s$ is the average standalone intrinsic value across all gap bidders. Since $V^s(z)$ is increasing in z, it holds that $\overline{V}_G^s = \int_0^{z^*} V^s(z) dF_Z(z)$ is smaller than $\overline{V}^s = \int_0^{z^{max}} V^s(z) dF_Z(z)$, provided that there exist firms with $z > z^*$. For notational convenience, define

$$\tilde{V}_G^s \equiv p_T \cdot \overline{V}_G^s + (1 - p_T) \cdot \tilde{V}^s. \tag{43}$$

It is straighforward to show that $\overline{V}_G^s < \tilde{V}_G^s < \overline{V}^s$.

Revaluations can then be expressed as the change in a gap bidder's standalone value relative to its pre-announcement value:

$$RV(z) = \frac{V_1^s(z) - V_0^s(z)}{V_0(z)}. \tag{44}$$

The revaluation component $RV(z)$ is negative for an average gap bidder:

$$\overline{RV} = (1 - p_A) \cdot \frac{\overline{V}_G^s - \tilde{V}^s}{\overline{V}_0}, \tag{45}$$

which is smaller than zero.

Furthermore, $RV(z)$ is decreasing in y. Taking the derivative of Equation (44) with respect to y yields

$$\begin{aligned} &\frac{d}{dy} RV(z) = -\frac{p_{AT} - p_A}{[V_0(z)]^2} \cdot \left[p_A \cdot A + (1 - p_A) \cdot V^{s\prime}(z) \cdot \overline{V}\right] - \frac{p_A \cdot V'(z)}{[V_0(z)]^2} \cdot B, \\ &\text{where } A = V^{s\prime}(z) \cdot V(z) - V'(z) \cdot V^s(z) \\ &\quad \text{and } B = (1 - p_A) \cdot \overline{V}^s - (1 - p_{AT}) \cdot \tilde{V}_G^s. \end{aligned} \tag{46}$$

It can be shown that $A > 0$, because $V^{s\prime}(z) > V'(z)$ and $V(z) > V^s(z)$ for $z \in (0, z^*]$. Similarly, $B > 0$ because $1 - p_A > 1 - p_{AT}$ and $\overline{V}^s > \tilde{V}_G^s$. Taken together, the derivative is positive; $RV(z)$ is decreasing in y. In other words, gap bidders experience more negative revaluations as the technology overlap between the target and the acquirer's industry rises.

Similar to Equation (16), which decomposes $AR(z)$ into a component driven by the announcement and one driven by the target choice, $RV(z)$ can also be decomposed into two components:

$$RV(z) = \underbrace{(1 - p_A) \cdot \frac{\tilde{V}_G^s - \overline{V}^s}{V_0(z)}}_{\text{driven by the announcement}} + \underbrace{(p_{AT} - p_A) \cdot \frac{V^s(z) - \tilde{V}_G^s}{V_0(z)}}_{\text{driven by the target choice}}. \tag{47}$$

The first component, $\tilde{V}_G^s - \overline{V}^s$, captures the shift in the market's expectation of standalone value upon learning that the firm is a gap bidder with probability p_T. The second component, $V^s(z) - \tilde{V}_G^s$, captures the market's updating of a gap bidder's standalone value upon learning its true technological position.

Synergies. In addition to revaluations, announcement returns also reflect acquisition synergies. I define synergies as the increase in a firm's intrinsic value when acquisitions are available:

$$S(z) = V(z) - V^s(z). \tag{48}$$

The announcement AR can then be decomposed as follows:

$$AR(z) = RV(z) + US(z), \tag{49}$$

$$US(z) = \underbrace{(1 - p_A) \cdot \frac{\tilde{S}_G - \overline{S}}{V_0(z)}}_{\text{driven by the announcement}} + \underbrace{(p_{AT} - p_A) \cdot \frac{S(z) - \tilde{S}_G}{V_0(z)}}_{\text{driven by the target choice}}, \tag{50}$$

where $US(z)$ denotes unexpected synergies, $\overline{S}$ denotes the average synergies across all firms (including non-gap bidders), $\overline{S}_G$ denotes the average synergies across all gap bidders, and $\tilde{S}_G \equiv p_T \cdot \overline{S}_G + (1 - p_T) \cdot \overline{S}$. Notably, announcement AR reflects only the component of synergies that is not anticipated by the market ex ante.

As with revaluations $RV(z)$, unexpected synergies $US(z)$ contribute to announcement ARs through two components. The first component, $\tilde{S}_G - \overline{S}$, captures the shift in the market's prior belief about expected synergies upon the announcement. The second component, $S(z) - \tilde{S}_G$, captures the market's further updating of this belief upon learning the gap bidder's true technological standing through the target choice.

A.1.4 Acquirer-target opacity difference and acquirer returns

This section formally derives the shape of $d_y AR(z)$ (Equation (18)) as a function of Δ (defined by Equation (19)). First, consider the scenario where the information observed by the market about the acquirer's and the target's technology is independent—the posterior probability of observing z follows $p_{AT} = 1 - (1 - p_A) \cdot (1 - p_T)$. Differentiating $d_y AR(z)$ with respect to Δ yields:

$$d_\Delta \left(d_y AR(z) \right) = -(1 - p_A) \cdot \frac{(1 - p_A) \cdot \left(\overline{V} - \overline{V}_G \right) + \overline{V}_G}{[V_0(z)]^2} \cdot c_y, \tag{51}$$

which is smaller than zero. Therefore, $d_y AR(z)$ is decreasing in Δ.

When the information observed about the acquirer and the target are nested, the market updates its belief following the rule: $p_{AT} = \max\{p_A, p_T\}$. Differentiating $d_y AR(z)$ with respect to Δ yields:

$$\frac{d}{d\Delta} \left(\frac{d}{dy} AR(z) \right) = \begin{cases} -\frac{p_A \cdot (1 - p_A) \cdot \left(\overline{V} - \overline{V}_G \right)}{[V_0(z)]^2} \cdot c_y & \text{if } p_A > p_T \\ -\frac{(1 - p_A) \cdot \left(\overline{V} - \overline{V}_G \right) + \overline{V}_G}{[V_0(z)]^2} \cdot c_y & \text{if } p_A \leq p_T \end{cases}. \tag{52}$$

The value of the function when $p_A > p_T$ is larger than when $p_A \leq p_T$. Therefore, although $d_y AR(z)$ is also decreasing in Δ, the slope is steeper when $p_A \leq p_T$.

A.2 Name matching

I establish record linkage between the universe of USTPO patent assignees and the universe of CapitalIQ companies.

First, I standardize all assignee and firm names and apply a frequency-based name matching algorithm following Hall, Jaffe, and Trajtenberg (2001).[18] For a second step, to improve matching quality while minimizing data losses, I require assignee-company pairs matched by the algorithm with non-identical standardized names to have a normalized Levenshtein distance of at least 0.5. All pairs below the threshold are excluded. The normalized Levenshtein distance is defined as the Levenshtein distance between two strings divided by the string length of the longer component. I further require each matched CapitalIQ firm to be assigned at least one patent within its sample period.

A.3 Construction procedures of the acquisition sample

Table A1: Sample construction. This table details the sample selection process and reports the number and percentage of observations remaining at each step. Public-target and private-target acquisitions are reported separately.

		Public-target		Private-target	
Step	Description	#	%	#	%
1	All transaction with a public acquirer and a public or private target announced in 1990-2020 and classified as "Acq. of assets," "Acq. Part. Int.," or "Acq. Maj. Int." from SDC via WRDS, with all public companies linked to CRSP and COMPUSTAT.	7,049	100.0%	19,586	100.0%
2	Keep deals where the acquirer seeks to purchase and own more than 50% of the target's shares after the acquisition.	5,605	79.5%	13,497	68.9%
3	Drop deals classified as recapitalization, repurchase or divestiture, and those with acquirers classified as an SPV.	5,413	76.8%	13,293	67.9%
4	Drop deals announced by the same acquirer within a one-month window of each other.	4,938	70.1%	11,397	58.2%
5	Merge all sample firms with patent data and all public firms with financial information. Exclude acquisitions with missing variables of interest, as well as those where the target firm has no patent records.	1,044	14.8%	962	4.9%

[18]The name standardization routine and the name matching algorithm can be downloaded under the following link: https://sites.google.com/site/patentdataproject/Home.

A.4 Variable definitions

Table A2: Variable definitions. This table defines the main variables used in this paper. All other variables are defined in the respective captions of the tables using them.

Variable	Definition	Data source
CAR(-1,1)	Cumulative abnormal returns for acquirer or targets from day -1 to day 1, with day 0 being the date of the acquisition announcement. See Section 3.2.3.	CRSP, SDC
CAR(-3,3)	Cumulative abnormal returns for acquirer or targets from day -3 to day 3, with day 0 being the date of the acquisition announcement. See Section 3.2.3.	CRSP, SDC
Total asset	Total book asset of a company. For private targets, this variable is imputed with SDC data in the following order: target asset, enterprise value, and ranking value.	COMPUSTAT, SDC
Log asset	Log-transformed total asset.	COMPUSTAT, SDC
Patent stock	The number of granted patents filed by a company in the past ten years.	USTPO
Log patent stock	Log-transformed patent stock.	USTPO
Tobin's Q	(Market capitalization + total asset - book equity) / total asset	COMPUSTAT
Log Tobin's Q	Log-transformed Tobin's Q.	COMPUSTAT
Market leverage	(Long-term debt + Long-term debt due within a year) / (Market capitalization + total asset - book equity)	COMPUSTAT
GapSignal	A measure for the acquirer's revealed technological gap. See Section 3.2.2.	USTPO, CapitalIQ
Alignment	A measure for the acquirer's technological alignment with its industry peers. See Section 3.2.2.	USTPO, CapitalIQ
% in cash	The percentage of the acquisition payment made in cash.	SDC
Tender offer	An indicator for whether an acquisition is a tender offer.	SDC
Hostile	An indicator for whether an acquisition is hostile.	SDC
Competitive	An indicator for whether an acquisition is subject to competitive bidding.	SDC
Toehold	The acquirer's ownership percentage in the target firm prior to the acquisition announcement.	SDC
Cross-border	An indicator for whether an acquisition involves two firms with residence in different countries.	SDC
Horizontal	An indicator for whether an acquisition involves two firms in the same 2-digit SIC industry.	SDC
Rumored deal	An indicator for whether an acquisition was rumored before the official announcement.	SDC
Prod. market rival	An indicator for whether an acquisition involves two firm in the same product market, as defined by Hoberg and Phillips (2010).	Hoberg and Phillips Data Library
Bid-ask spread	The annual effective bid-ask spread as defined by Abdi and Ranaldo (2017).	CRSP
Analyst following	The number of analysts following a company over a year.	I/B/E/S

Table A2: Variable definitions (continued).

Variable	Definition	Data source
Transparent target	An indicator equal to 1 if a private target is regarded as exceptionally transparent, and 0 otherwise. A private target is regarded as transparent if it meets all of the following conditions: (a) it has been assigned a SEDOL code prior to the announcement date; (b) it is recorded as having issued a corporate bond in FINRA data; and (c) CapitalIQ headlines document the bond issuance.	SDC, FINRA, CapitalIQ
R&D/asset	R&D expenses / total asset.	COMPUSTAT
ΔR&D/asset	Year-over-year change in R&D/asset.	COMPUSTAT
Patent index	An index reflecting a firm's technological standing, as defined by Bena and Li (2014).	USTPO
ΔPatent index	Year-over-year change in patent index.	USTPO
Age	The number of years since the company first appeared in the COMPUSTAT database.	COMPUSTAT
Productivity	Total factor productivity, estimated as the residual from regressing log-transformed sales against log-transformed PP&E and log-transformed employee count over 10-year rolling windows for each 2-digit SIC industry.	COMPUSTAT
Cash flow/asset	Net cashflow from operating activities / year-beginning total asset.	COMPUSTAT
Cash/asset	Cash holdings / total asset.	COMPUSTAT
Capex/asset	Capital expenditure / total asset.	COMPUSTAT
Sales growth	Year-over-year change in sales scaled by the average sales over the two years.	COMPUSTAT
Competition	One minus the Herfindahl index of sales across firms within the same 2-digit SIC industry.	COMPUSTAT
Tech competition	One minus the Herfindahl index of patent stock across firms within the same 2-digit SIC industry.	USTPO
Female executive	An indicator for whether a company has a female executive.	Execucomp
Female CEO	An indicator for whether a company has a female CEO.	Execucomp

A.5 Revaluations and relative opacity

This appendix presents the estimation strategies for the function $g(\cdot)$, which captures how relative acquirer-target opacity moderates revaluation intensity—that is, the marginal effect of the acquirer's revealed technological gap on its announcement returns. See Section 4.2 for more details. For a theoretical discussion motivating this relationship, see Section 2.3.3.

The first specification models $g(\cdot)$ as a piecewise linear function that allows for a potential cusp at $\Omega = 0$. This is defined as:

$$g(\Omega) = \delta_0 + \delta_1 \Omega + \delta_2 (\Omega)_+, \tag{53}$$

where $(\cdot)_+$ denotes the positive-part operator, which equals the input if positive and zero otherwise. This specification allows for a discrete change in slope at $\Omega = 0$.

The second specification models $g(\cdot)$ using a restricted cubic spline (RCS), which provides a smooth and flexible approximation to potential nonlinearities. Following Harrell (2001). I employ a basis function approach with five knots. The k-th knot, t_k, is placed at the $100 \cdot k/(k+1)$ percentile of the distribution of Ω. The RCS representation of $g(\cdot)$ is:

$$g(\cdot) = \psi_0 + \psi_1 F_1 + \psi_2 F_2 + \psi_3 F_3 + \psi_4 F_4, \tag{54}$$

where $F_1 = \Omega$ and for $j = 1, 2, 3$:

$$F_{j+1} = (\Omega - t_j)_+^3 - \frac{t_k - t_j}{t_k - t_{k-1}}(\Omega - t_{k-1})_+^3 + \frac{t_{k-1} - t_j}{t_k - t_{k-1}}(\Omega - t_k)_+^3. \tag{55}$$

This construction ensures smoothness and continuity at the knot points and imposes linearity in the tails.

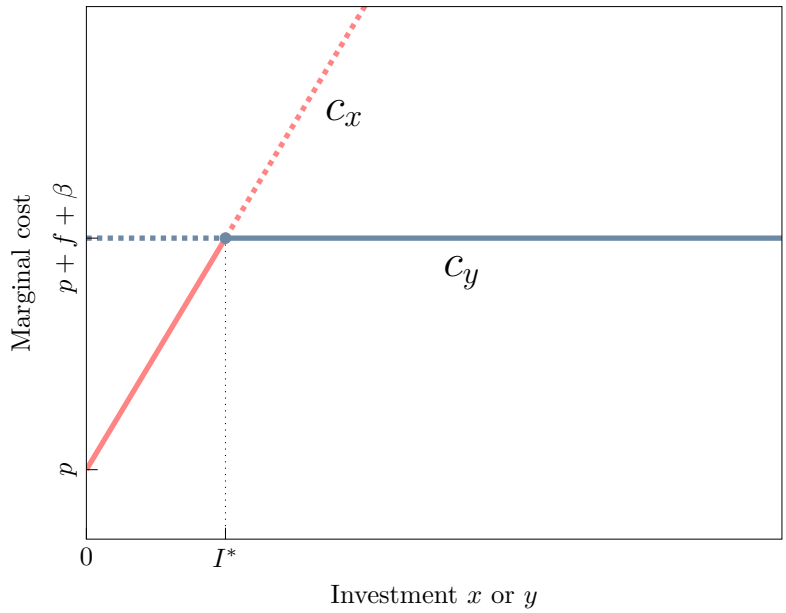
Both specifications—piecewise linear and RCS—can be expressed as linear combinations of transformed versions of Ω, denoted $\{J^s\}_{s=1}^n$, along with a constant. I estimate the following regression model:

$$\begin{aligned} CAR(-\tau, \tau)_{id} = \eta_0 \cdot GapSignal_d + \sum_{s=1}^{n} \eta_s \cdot GapSignal_d \times J_d^s + \sum_{s=1}^{n} \theta_s \cdot J_d^s \\ + \mathbf{X}_{id} \cdot \mathbf{\Gamma} + \alpha_u + \alpha_v + \alpha_t + \varepsilon_{id}, \end{aligned} \tag{56}$$

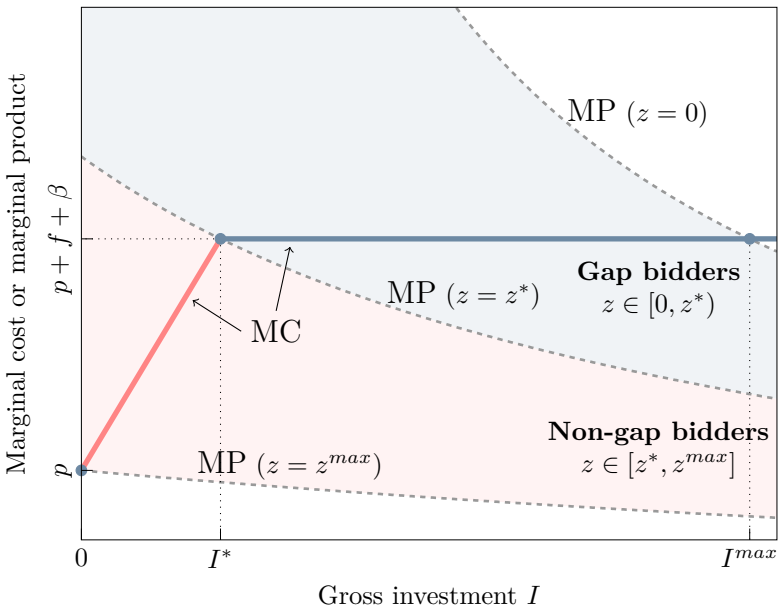
where i indexes acquirers, d indexes acquisition deals, u and v denote 2-digit SIC industry of the acquirer and the target, respectively, and t represents the calendar year. $\mathbf{X}_{id}$ is a vector of control variables capturing firm- and deal-level characteristics, as introduced in Section 3.2.3. The regression includes industry and year fixed effects α_u, α_v, α_t to account for unobserved heterogeneity across industries and time.

The model includes the transformed terms $\{J_d^s\}_{s=1}^n$ and their interactions with *GapSignal* as regressors. This specification identifies the function $g(\cdot)$ directly, since the partial derivative of the regression model with respect to *GapSignal* yields $g(\cdot)$.

B Figures

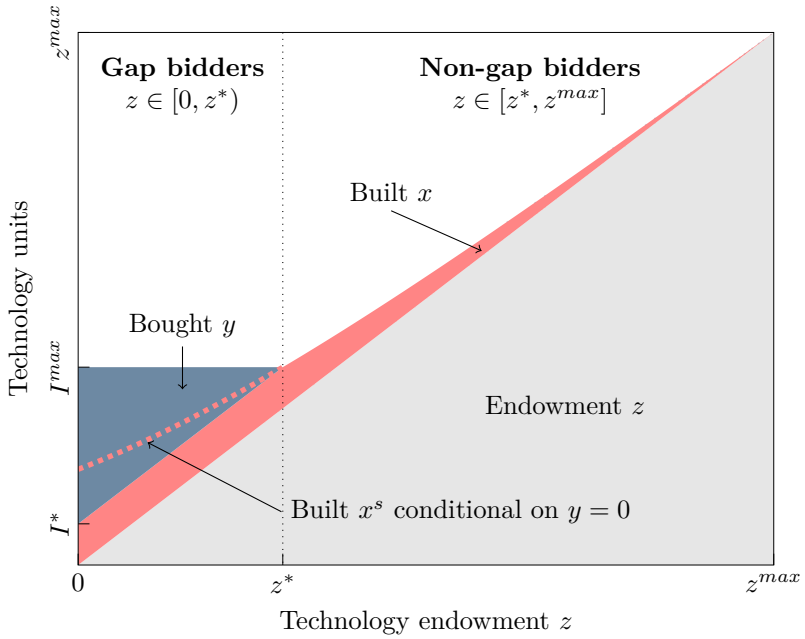


(a): The pecking order of technological investment

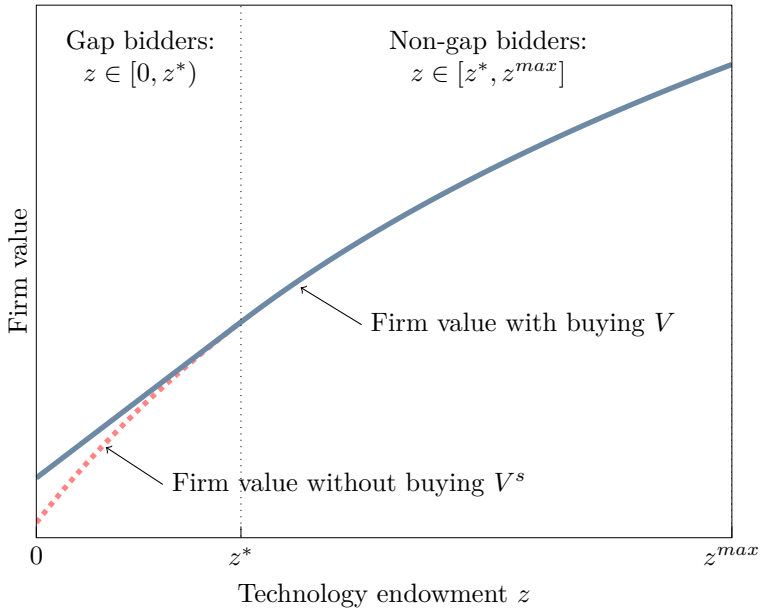


(b): The optimal investment level

Figure 1: The pecking-order investment strategy. The two panels plot the marginal product (MP) of technology (Equation (4)), the marginal cost (MC) of building (Equation (5)), the MC of buying (Equation (6)), and the MC of technology under the pecking order. The thresholds I^*, I^{max}, z^*, and z^{max} are defined by Equations (7), (34), (33), and (35), respectively. The parameters are calibrated as follows: $\kappa = 1$, $\alpha = 0.3$, $f = \beta = 0.025$, $p = 0.05$, and $\phi = 2$.



(a): The optimal investment strategy



(b): Firm value under the optimal strategy

Figure 2: The optimal investment strategy and firm value. Panel (a) plots the optimal investment through in-house development $x(z)$ (Equation (36)) and acquisitions $y(z)$ (Equation (37)), respectively, and that absent acquisitions $x^s(z)$ (Equation (38)) as a function of technology endowment z. Panel (b) plots the firm value under the optimal strategy $V(z)$ (Equation (39)) and that absent acquisitions $V^s(z)$ (Equation (40)) as a function of z. The thresholds I^*, I^{max}, z^*, and z^{max} are defined by Equations (7), (34), (33), and (35), respectively. The parameters are calibrated as follows: $\kappa = 1$, $\alpha = 0.3$, $f = \beta = 0.025$, $p = 0.05$, and $\phi = 2$.

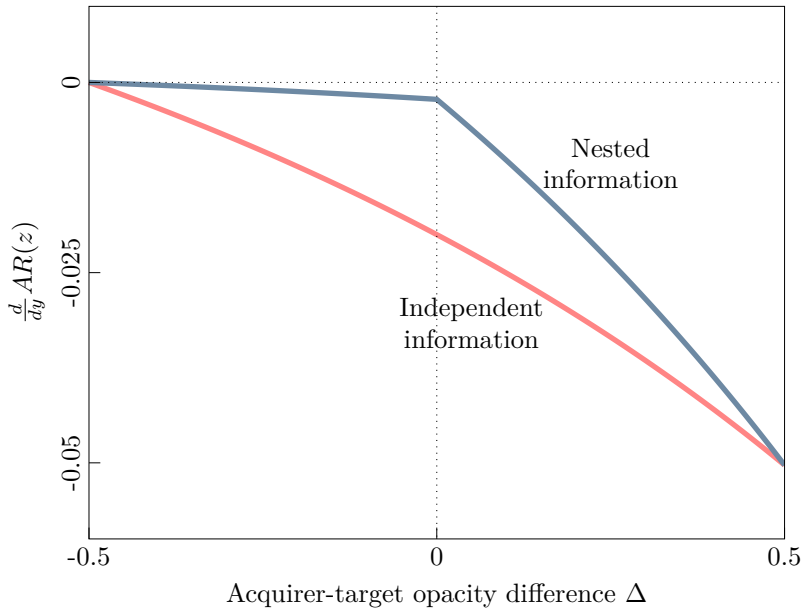
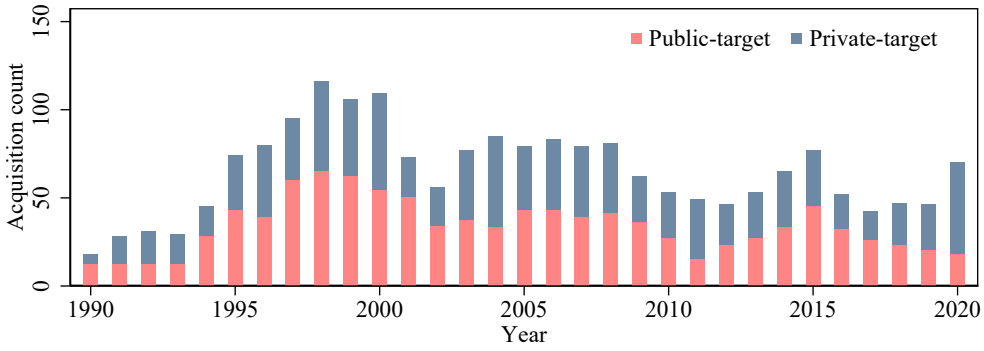
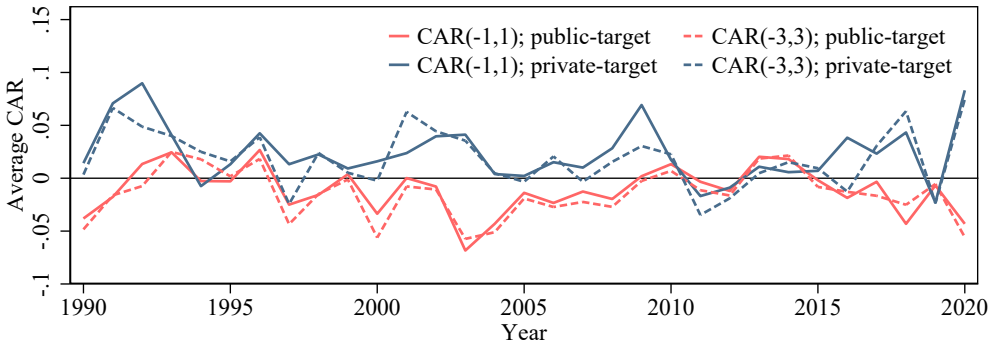


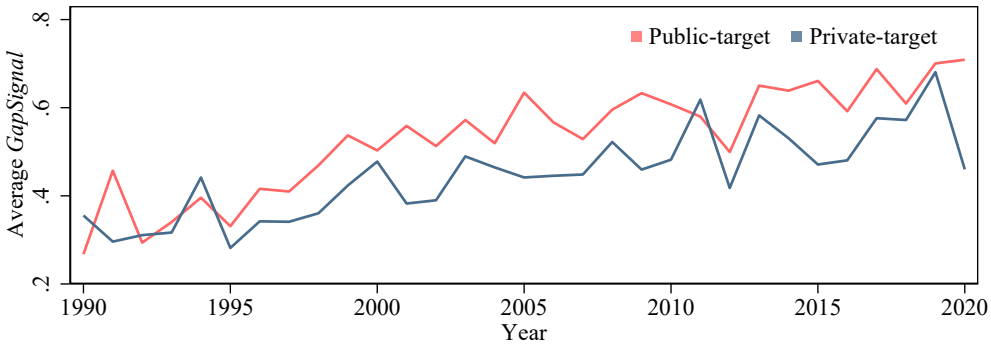
Figure 3: Firm opacity as a moderator. The figure visualizes Equation (18), the marginal effect of the number of technology units bought from the target (y) on acquirer returns ($AR(z)$) as a function of Δ, the acquirer-target opacity difference (Equation (19)). The red solid line assumes the market's observations of the acquirer's and the target's technology are independent. The blue line assumes the information from these two observations is nested. The parameters are calibrated as follows: $p_A = 0.5$, $\overline{V} = 1.7$, $\overline{V}_G = 1.3$, $c_y = p + f + \beta = 0.1$.



(a): Sample distribution by year

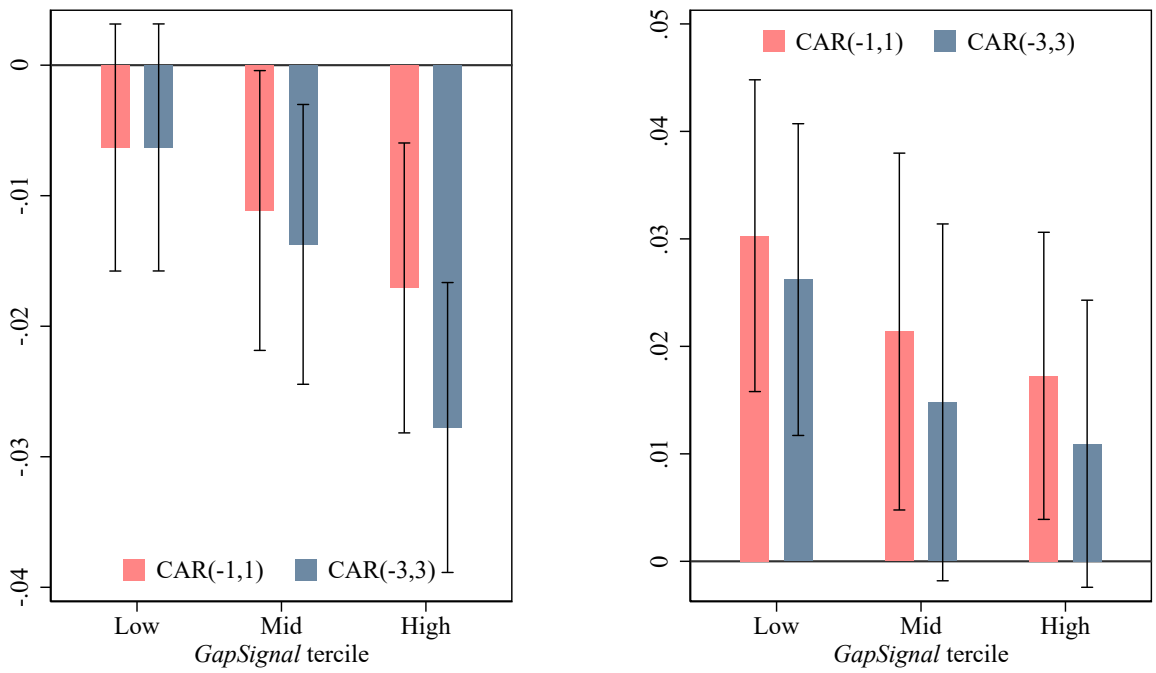


(b): Average acquirer CARs by year



(c): Average *GapSignal* by year

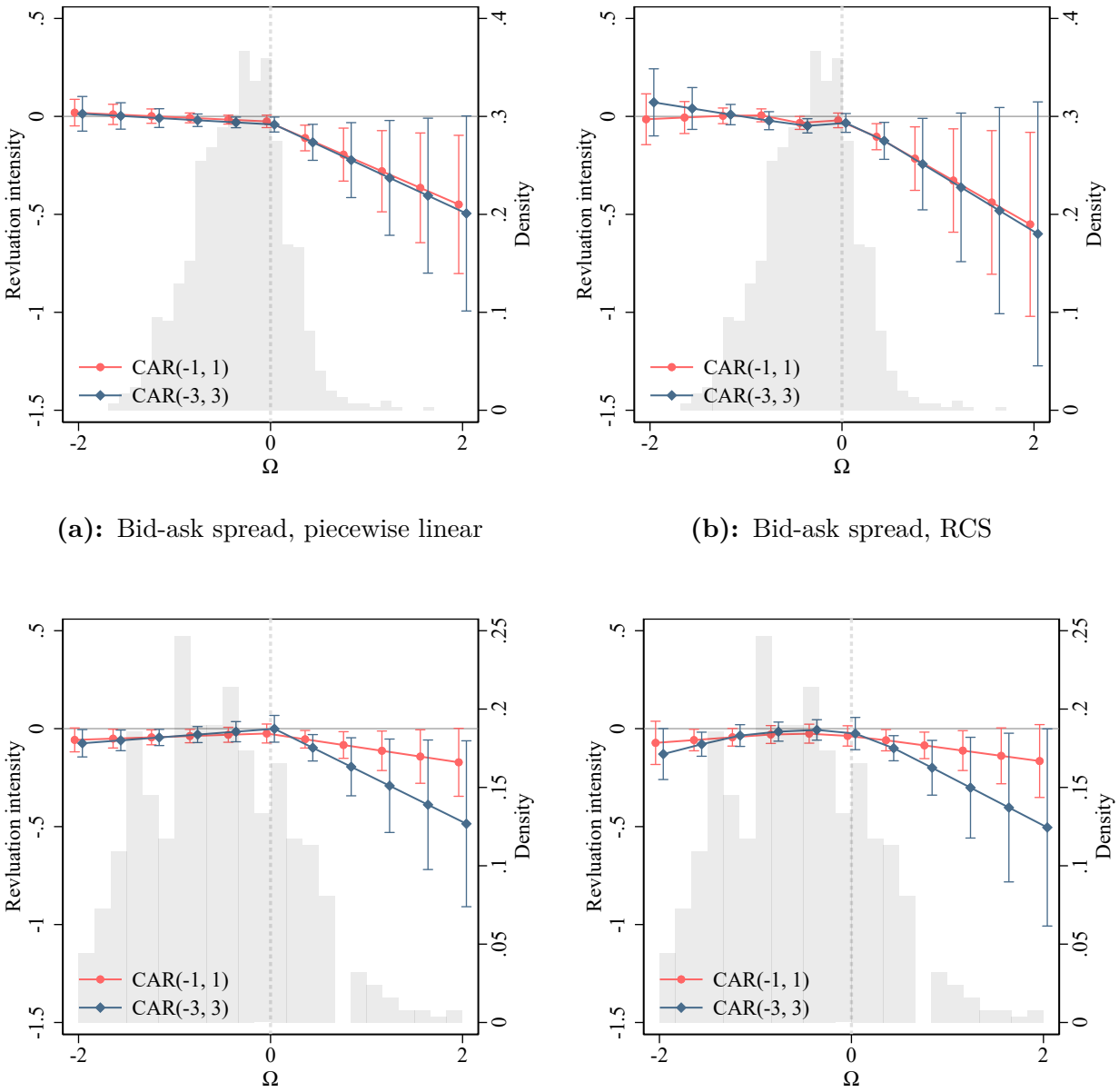
Figure 4: Acquisition sample by year. This figure summarizes the annual distribution and characteristics of the acquisition sample by the target's listing status. Panel (a) shows the number of public- and private-target acquisitions by year. Panel (b) reports average acquirer CARs over two event windows—CAR(-1,1) and CAR(-3,3)—by year and target type. Panel (c) presents the average *GapSignal* score by year, separately for public- and private-target acquisitions.



(a): Public-target

(b): Private-target

Figure 5: Revealed technological gaps and acquirer returns. This figure sorts sample acquirers of public and private targets separately into terciles by *GapSignal*, and reports the average acquirer CARs over two event windows—CAR(-1,1) and CAR(-3,3)—within each tercile, along with 95% confidence intervals.



(a): Bid-ask spread, piecewise linear

(b): Bid-ask spread, RCS

(c): Analyst following, piecewise linear

(d): Analyst following, RCS

Figure 6: The moderating effect of relative opacity. This figure plots the estimated function $g(\cdot)$, as defined in Equation (30), which captures how revaluation intensity—the marginal effect of *GapSignal* on acquirer CARs—varies with acquirer-target relative opacity Ω. Panels (a) and (b) use bid-ask spreads as the proxy for opacity, while Panels (c) and (d) use (negative) analyst following. Panels (a) and (c) estimate $g(\cdot)$ using a piecewise linear specification with a breakpoint at $\Omega = 0$; Panels (b) and (d) use a RCS specification. Each panel reports point estimates of $g(\cdot)$ across selected values of Ω, along with 95% confidence intervals. The background histograms depict the distribution of Ω in the estimation sample.

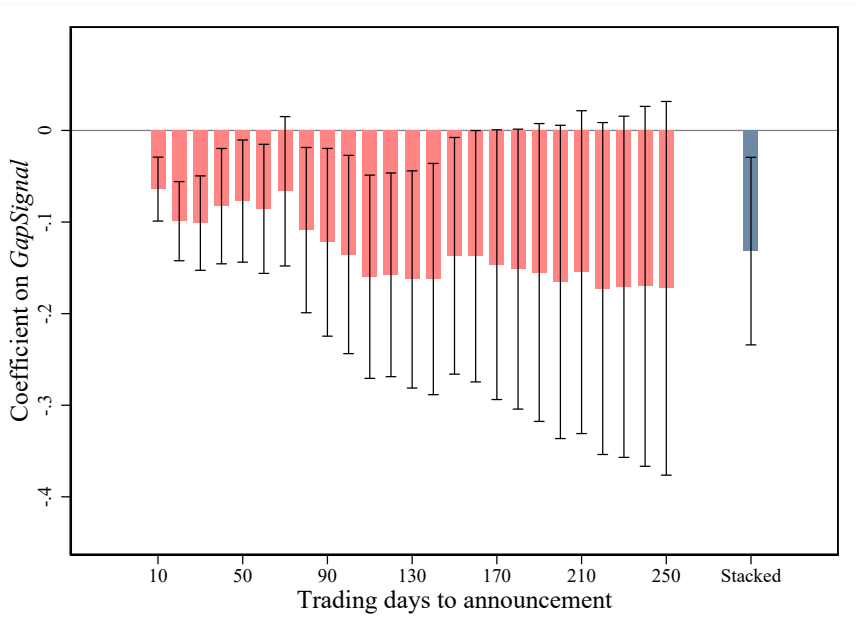


Figure 7: Long-term effect. This figure plots the coefficient estimates on *GapSignal* of Equation (32) for each horizon $h \in \{10, 20, \ldots, 250\}$ and all horizons stacked. The stacked regression additionally controls for horizon fixed effects and clusters standard errors at the transaction level. The sample firms are public target-acquirers.

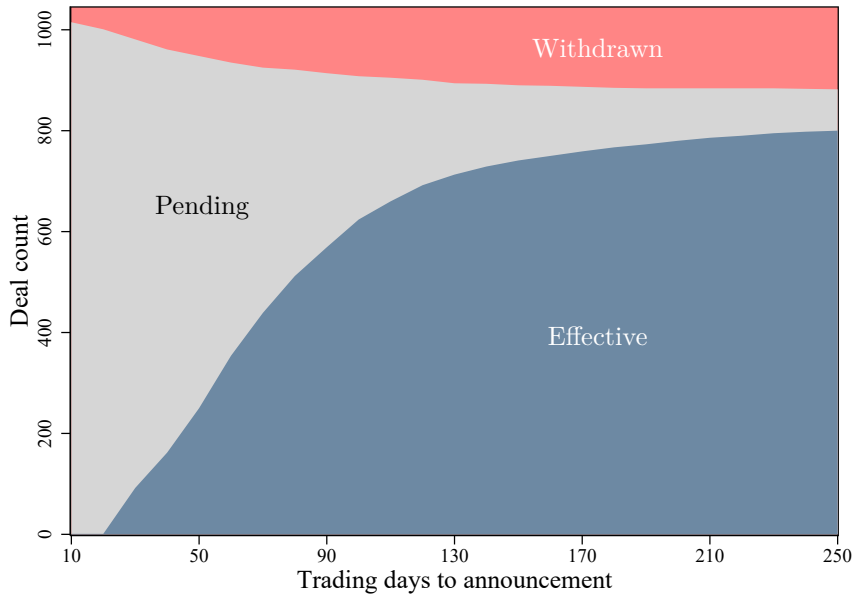


Figure 8: The evolution of deal status. This figure demonstrates the evolution of the sample deals with public targets up to 250 trading days after the acquisition announcement. Firms are classified into three categories: effective, pending, and withdrawn.

C Tables

Table 1: Summary statistics. This table presents summary statistics for the full sample of acquisitions, separated by whether the target is a public or private firm. Panel A reports characteristics of the acquirers, including announcement-window cumulative abnormal returns (CARs), size, patenting activity, Tobin's Q, and market leverage. Panel B reports analogous statistics for targets. Panel C summarizes deal-level characteristics, including the *GapSignal* and *Alignment* scores, method of payment, deal type, and competitive environment. The last column reports the difference in means between public- and private-target deals. All variables are defined in Table A2.*, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Public-target (N = 1,044)			Private-target (N = 962)			Mean difference
	Mean	Median	SD	Mean	Median	SD	
Panel A: Acquirer characteristics							
CAR(-1,1)	-0.012	-0.005	0.083	0.023	0.006	0.125	-0.035***
CAR(-3,3)	-0.016	-0.010	0.099	0.017	0.007	0.136	-0.033***
Total asset ($bn)	16.418	1.706	66.826	3.738	0.343	13.393	12.680***
Log asset	7.503	7.442	2.294	5.946	5.837	2.118	1.558***
Patent stock	263.430	1.000	1,438.176	165.400	1.000	989.543	98.030**
Log patent stock	2.042	0.693	2.472	1.796	0.693	2.179	0.246***
Tobin's Q	2.765	2.037	2.374	3.049	2.098	2.719	-0.284***
Log Tobin's Q	0.798	0.711	0.610	0.858	0.741	0.677	-0.060**
Market leverage	0.107	0.075	0.115	0.076	0.027	0.103	0.030***
Panel B: Target characteristics							
CAR(-1,1)	0.249	0.187	0.315				
CAR(-3,3)	0.255	0.205	0.384				
Total asset ($bn)	7.672	0.203	82.354	0.246	0.038	1.297	7.427***
Log asset	5.632	5.311	2.141	3.723	3.646	1.822	1.909***
Patent stock	108.470	9.000	850.381	8.615	3.000	19.958	99.855***
Log patent stock	2.615	2.303	1.641	1.621	1.386	0.951	0.993***
Tobin's Q	2.429	1.713	2.245				
Log Tobin's Q	0.646	0.538	0.638				
Market leverage	0.105	0.040	0.135				
Panel C: Deal characteristics							
GapSignal	0.536	0.579	0.302	0.451	0.442	0.304	0.085***
Alignment	0.319	0.089	0.359	0.316	0.126	0.355	0.003
% in cash	0.455	0.331	0.460	0.361	0.000	0.427	0.094***
Tender offer	0.223	0.000	0.417	0.004	0.000	0.064	0.219***
Hostile	0.022	0.000	0.147	0.001	0.000	0.032	0.021***
Competitive	0.071	0.000	0.257	0.001	0.000	0.032	0.070***
Toehold	0.007	0.000	0.041	0.010	0.000	0.052	-0.003*
Cross-border	0.154	0.000	0.361	0.214	0.000	0.410	-0.060***
Horizontal	0.680	1.000	0.467	0.543	1.000	0.498	0.138***
Prod. market rival	0.563	1.000	0.496				
Rumored deal	0.127	0.000	0.334	0.013	0.000	0.111	0.115***
Completed	0.785	1.000	0.411	0.917	1.000	0.276	-0.132***

Table 2: Public-target acquirers and revealed gaps. This table reports the regression results of Equation (28) based on a sample of public-target acquisitions. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) in Panel A and CAR(-3,3) in Panel B. The main independent variable of interest is *GapSignal*, which captures the acquirer's technological gap revealed by the acquisition. Only a subset of control variables is shown for brevity; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: CAR(-1,1)

Dep. var.:	CAR(-1,1) of public-target acquirers				
	(1)	(2)	(3)	(4)	(5)
GapSignal	-0.028***	-0.030***	-0.029**	-0.059***	-0.057***
	(-2.85)	(-2.73)	(-2.54)	(-3.18)	(-3.00)
Alignment		-0.016**	-0.012	-0.009	-0.004
		(-1.97)	(-0.97)	(-0.50)	(-0.21)
% in cash		0.026***	0.018**	0.029***	0.026***
		(3.84)	(2.53)	(2.87)	(2.63)
Log asset (A)			0.005**	0.009***	0.009***
			(2.47)	(3.03)	(2.94)
Log asset (T)			-0.002	-0.008**	-0.007**
			(-0.76)	(-2.19)	(-2.04)
Deal controls	No	Yes	Yes	Yes	Yes
Firm controls	No	No	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	Yes	Yes	No	No
AInd × Year, Tind × Year FE	No	No	No	Yes	No
AInd × Tind × Year FE	No	No	No	No	Yes
Adj. R^2	0.050	0.074	0.080	0.085	0.113
N	1,044	1,044	1,044	757	650

Panel B: CAR(-3,3)

Dep. var.:	CAR(-3,3) of public-target acquirers				
	(1)	(2)	(3)	(4)	(5)
GapSignal	-0.041***	-0.043***	-0.044***	-0.058***	-0.061***
	(-3.36)	(-3.15)	(-3.16)	(-2.63)	(-2.72)
Alignment		-0.010	-0.007	-0.006	-0.005
		(-0.99)	(-0.43)	(-0.27)	(-0.21)
% in cash		0.027***	0.018**	0.035***	0.034***
		(3.42)	(2.18)	(2.96)	(2.84)
Log asset (A)			0.006***	0.011***	0.012***
			(2.61)	(3.27)	(3.64)
Log asset (T)			-0.005*	-0.011***	-0.010**
			(-1.76)	(-2.71)	(-2.45)
Deal controls	No	Yes	Yes	Yes	Yes
Firm controls	No	No	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	Yes	Yes	No	No
AInd × Year, Tind × Year FE	No	No	No	Yes	No
AInd × Tind × Year FE	No	No	No	No	Yes
Adj. R^2	0.044	0.052	0.055	0.053	0.083
N	1,044	1,044	1,044	757	650

Table 3: What if the gap is known? This table replicates Columns (3)–(5) in both panels of Table 2, while interacting *GapSigal* with three variables. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) in Panel A and CAR(-3,3) in Panel B. The main independent variable of interest is *GapSignal*, which captures the acquirer's technological gap revealed by the acquisition. *Toehold* is the percentage pre-bid target share owned by the acquirer. *Rumored deal* is an indicator that equals one if a deal is rumored, and zero otherwise. *Back-to-back* is an indicator if an acquisition is preceded by another acquisition initiated by the same acquirer within 50 calendar days. Control variables are omitted; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Bidders with toeholds

Sample:	Acquirers of public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal × Toehold	0.073	0.535***	0.721***	0.104	0.583*	0.830**
	(0.57)	(2.91)	(3.21)	(0.58)	(1.96)	(2.28)
GapSignal	-0.030**	-0.061***	-0.060***	-0.045***	-0.060***	-0.064***
	(-2.58)	(-3.32)	(-3.17)	(-3.21)	(-2.78)	(-2.90)
Toehold	-0.010	-0.445***	-0.593***	-0.013	-0.511**	-0.732**
	(-0.10)	(-2.95)	(-3.18)	(-0.09)	(-2.02)	(-2.35)
Deal, firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.079	0.089	0.118	0.055	0.056	0.088
N	1,044	757	650	1,044	757	650

Panel B: Rumored deals

Sample:	Acquirers of public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal × Rumored deal	0.049**	0.044	0.025	0.071***	0.080*	0.061
	(2.32)	(1.08)	(0.60)	(2.59)	(1.82)	(1.45)
GapSignal	-0.035***	-0.063***	-0.060***	-0.053***	-0.065***	-0.067***
	(-2.89)	(-3.21)	(-2.94)	(-3.56)	(-2.81)	(-2.81)
Rumored deal	-0.017	-0.006	0.013	-0.033	-0.039	-0.022
	(-1.06)	(-0.18)	(0.40)	(-1.59)	(-1.20)	(-0.71)
Deal, firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.083	0.085	0.111	0.059	0.055	0.083
N	1,044	757	650	1,044	757	650

Table 3: What if the gap is known (continued)?

Panel C: Back-to-back acquisitions

Sample:	Acquirers of public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal × Back-to-back	0.080	0.148**	0.183***	0.071	0.151**	0.170**
	(1.49)	(2.00)	(2.59)	(1.33)	(2.00)	(2.34)
GapSignal	-0.032***	-0.065***	-0.066***	-0.046***	-0.063***	-0.068***
	(-2.83)	(-3.61)	(-3.63)	(-3.33)	(-2.94)	(-3.13)
Back-to-back	-0.055	-0.135**	-0.162***	-0.039	-0.122**	-0.138**
	(-1.42)	(-2.55)	(-3.24)	(-1.00)	(-2.12)	(-2.45)
Deal, firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.081	0.096	0.128	0.054	0.057	0.088
N	1,044	757	650	1,044	757	650

Table 4: Acquirer returns and revealed gaps with private targets. This table reports the regression results of Equation (28) based on a sample of private-target acquisitions. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) in Panel A and CAR(-3,3) in Panel B. The main independent variable of interest is *GapSignal*, which captures the acquirer's technological gap revealed by the acquisition. Only a subset of control variables is shown for brevity; a complete list is provided in Section 3.2.3. $AInd$ and $TInd$ denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: CAR(-1,1)

Dep. var.:	CAR(-1,1) of private-target acquirers				
	(1)	(2)	(3)	(4)	(5)
GapSignal	-0.012	0.001	0.010	0.035	0.028
	(-0.73)	(0.07)	(0.60)	(1.40)	(1.07)
Alignment		-0.031**	-0.045**	-0.031	-0.037
		(-2.58)	(-2.52)	(-1.19)	(-1.39)
% in cash		-0.016	-0.008	-0.016	-0.014
		(-1.64)	(-0.84)	(-1.24)	(-1.07)
Log asset (A)			-0.013***	-0.009**	-0.011**
			(-4.59)	(-2.48)	(-2.58)
Log asset (T)			0.003	-0.002	0.000
			(1.03)	(-0.40)	(0.10)
Deal controls	No	Yes	Yes	Yes	Yes
Firm controls	No	No	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	Yes	Yes	No	No
AInd × Year, Tind × Year FE	No	No	No	Yes	No
AInd × Tind × Year FE	No	No	No	No	Yes
Adj. R^2	0.114	0.124	0.149	0.048	0.021
N	962	962	962	711	559

Panel B: CAR(-3,3)

Dep. var.:	CAR(-3,3) of private-target acquirers				
	(1)	(2)	(3)	(4)	(5)
GapSignal	-0.012	0.005	0.008	0.002	-0.005
	(-0.71)	(0.29)	(0.43)	(0.05)	(-0.16)
Alignment		-0.048***	-0.062***	-0.044	-0.033
		(-3.61)	(-3.11)	(-1.46)	(-1.05)
% in cash		-0.015	-0.009	-0.030*	-0.025
		(-1.26)	(-0.79)	(-1.69)	(-1.53)
Log asset (A)			0.004	-0.002	0.000
			(1.30)	(-0.39)	(0.04)
Log asset (T)			0.004	0.001	0.000
			(1.41)	(0.14)	(-0.01)
Deal controls	No	Yes	Yes	Yes	Yes
Firm controls	No	No	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	Yes	Yes	No	No
AInd × Year, Tind × Year FE	No	No	No	Yes	No
AInd × Tind × Year FE	No	No	No	No	Yes
Adj. R^2	0.051	0.064	0.071	-0.041	0.017
N	962	962	962	711	559

Table 5: Transparent private targets. This table replicates Table 4, while interacting *GapSignal*—the proxy for the acquirer's revealed technological gap—with an indicator for exceptionally transparent targets. *Late-stage target* in Panel A equals 1 if the target received a late-stage VC funding round within five years prior to the acquisition announcement, and 0 otherwise. *Bond-issuing target* in Panel B equals 1 if has issued corporate bonds or other securities prior to the acquisition announcement prior to the acquisition announcement, and 0 otherwise. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) and CAR(-3,3). Control variables are omitted for brevity; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: VC-backed late-stage private targets

Sample:	Acquirers of private targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal × Late-stage target	-0.080*	-0.125**	-0.142*	-0.076*	-0.120**	-0.127*
	(-1.76)	(-1.97)	(-1.80)	(-1.73)	(-2.01)	(-1.81)
GapSignal	0.017	0.046*	0.043	0.015	0.011	0.006
	(0.95)	(1.79)	(1.57)	(0.67)	(0.30)	(0.16)
Late-stage target	0.062*	0.098**	0.102*	0.062**	0.104**	0.100**
	(1.89)	(2.06)	(1.72)	(2.07)	(2.46)	(1.98)
Deal, firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.152	0.058	0.031	0.073	-0.033	0.021
N	962	711	559	962	711	559

Panel B: Bond-issuing private targets

Sample:	Acquirers of private targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal × Bond-issuing target	-0.193***	-0.160	-0.269**	-0.206***	-0.204	-0.314***
	(-2.63)	(-1.40)	(-2.53)	(-2.70)	(-1.64)	(-2.60)
GapSignal	0.021	0.040	0.038	0.020	0.008	0.006
	(1.24)	(1.61)	(1.43)	(1.00)	(0.25)	(0.16)
Bond-issuing target	0.090*	0.046	0.135*	0.086*	0.051	0.126
	(1.84)	(0.54)	(1.77)	(1.77)	(0.60)	(1.55)
Deal, firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.157	0.055	0.033	0.079	-0.030	0.034
N	962	711	559	962	711	559

Table 6: Predicting gap bidders. This table presents estimation results of Equation (31). The dependent variable *GapSignal* measures the acquirer's technological gap revealed by the target's technological profile. The dependent variables are acquisition antecedents discussed in Section 4.3.1. $AInd$ and $TInd$ denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Sample:	Acquirers of public targets							
Dep. var.:	*GapSignal*							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
R&D/asset	0.249***							0.241***
	(5.48)							(4.81)
ΔR&D/asset	-0.147***							-0.162***
	(-3.46)							(-3.58)
Patent index	0.000							0.000
	(0.49)							(-0.64)
ΔPatent index	0.0001*							0.000
	(1.67)							(1.42)
Log asset		0.013***						0.019***
		(2.68)						(3.52)
Log patent stock		0.006						0.003
		(1.63)						(0.89)
Age		-0.002***						-0.002**
		(-2.73)						(-2.21)
Log Tobin's Q			0.050***					0.038**
			(3.60)					(2.39)
Productivity			0.031**					0.031**
			(2.26)					(2.05)
Cash flow/asset				-0.011				-0.053**
				(-0.55)				(-2.50)
Cash/asset				0.035				-0.001
				(0.66)				(-0.02)
Market leverage				-0.184**				-0.116
				(-1.97)				(-1.13)
Capex/asset					-0.207			-0.269
					(-1.01)			(-1.30)
Sales growth					-0.020			-0.064***
					(-0.86)			(-2.82)
Competition						-0.843***		-0.835***
						(-2.66)		(-2.67)
Tech competition						0.146		0.173
						(1.32)		(1.56)
Female executive							0.033	0.022
							(1.55)	(1.03)
Female CEO							0.010	0.014
							(0.20)	(0.28)
Year, AInd, TInd FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.370	0.367	0.369	0.361	0.358	0.361	0.359	0.390
N	1,044	1,044	1,044	1,044	1,044	1,044	1,044	1,044

Table 7: Prediction or suprise? This table replicates Columns (3) to (5) in Table 2, Panel A and Panel B, while decomposing *GapSignal*—the proxy for the acquirer's revealed technological gap—into a fitted value that is predicted by acquisition antecedents discussed in Section 4.3.1, and a residual that capture information orthogonal to observable acquirer characteristics. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) and CAR(-3,3). Control variables are omitted for brevity; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Sample:	Acquirers of public targets					
Dep. var.:	CAR(-1,1)			CAR(-3,3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal (fitted)	-0.017	-0.082	-0.071	-0.071	-0.111	-0.102
	(-0.21)	(-0.76)	(-0.64)	(-1.03)	(-1.17)	(-1.09)
GapSignal (residual)	-0.030**	-0.058***	-0.057***	-0.043***	-0.055**	-0.059***
	(-2.54)	(-3.12)	(-2.97)	(-3.04)	(-2.51)	(-2.65)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.079	0.084	0.111	0.055	0.052	0.081
N	1,044	757	650	1,044	757	650

Table 8: Revaluation of industry peers. This table reports regression results based on a sample of acquirers' industry peers. The dependent variables are the peers' CARs over two event windows, CAR(-1,1) and CAR(-3,3), in Panel A, and two indicators, *Target in 3 years* and *Acquirer in 3 years* in Panel B. They equal one if the firm becomes an acquisition target or acquirer in the next three years, respectively. The main independent variable of interest is *GapSignal*, which captures the acquirer's technological gap revealed by the acquisition. Control variables include deal-level characteristics listed in Section 3.2.3 and the peers' *Log size*, *Log Tobin's Q*, *Market leverage*, and *Log patent stock*. *Ind* denotes the 2-digit SIC industries of the peer firm. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are clustered at the deal level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Market reactions

Sample:	Peers of public-target acquirers			
Dep. var.:	CAR(-1, 1)		CAR(-3, 3)	
	(1)	(2)	(3)	(4)
GapSignal	0.005***	0.005**	0.003	0.002
	(2.67)	(2.25)	(1.07)	(0.57)
Firm, deal controls	Yes	Yes	Yes	Yes
Year, Ind FE	Yes	No	Yes	No
AInd × Year FE	No	Yes	No	Yes
Adj. R^2	0.002	0.003	0.003	0.005
N	100,663	100,659	100,663	100,659

Panel B: Future acquisition involvement

Sample:	Peers of public-target acquirers			
Dep. var.:	Target in 3 years		Acquirer in 3 years	
	(1)	(2)	(3)	(4)
GapSignal	-0.002	0.000	-0.002	0.000
	(-1.33)	(0.95)	(-0.53)	(0.38)
Firm, deal controls	Yes	Yes	Yes	Yes
Year, Ind FE	Yes	No	Yes	No
AInd × Year FE	No	Yes	No	Yes
Adj. R^2	0.006	0.011	0.050	0.055
N	100,663	100,659	100,663	100,659

Table 9: Deal status. The dependent variable *GapSignal* proxies for the acquirer's revealed technological gap. The dependent variables are public-target acquirer's CARs over the window $[-3, t]$, with $t \in \{10, 20, ..., 250\}$. Only a subset of control variables are reported for brevity; a complete list is provided in Section 3.2.3. $AInd$ and $TInd$ denote the 2-digit SIC industries of the acquirer and the target, respectively. $Horizon$ denotes different event windows. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dep. var.:	CAR(-3, t) for acquirers of public targets			
Subsample:	All	Effective	Pending	Withdrawn
	(1)	(2)	(3)	(4)
GapSignal	-0.132**	-0.102	-0.100**	-0.492**
	(-2.52)	(-1.37)	(-2.46)	(-2.46)
Firm, deal controls	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	Yes	Yes	Yes
Horizon FE	Yes	Yes	Yes	Yes
Adj. R^2	0.151	0.154	0.115	0.740
N	25,786	14,108	8,500	3,176

Table 10: Overpayment? Panel A replicates Columns (3) to (5) in both panels of Table 2 for public targets. Panel B replaces the dependent variable with bid premia measured with two price benchmarks: closing prices 1-day and 1-week prior to the acquisition announcement. The dependent variable *GapSignal* proxies for the acquirer's revealed technological gap. The dependent variables are the target's CARs over two event windows: CAR(-1,1) and CAR(-3,3). Only a subset of control variables are reported in Panel A and all are omitted in Panel B; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Market reactions

Sample:	Public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal	-0.017	-0.082	-0.071	-0.071	-0.111	-0.102
	(-0.21)	(-0.76)	(-0.64)	(-1.03)	(-1.17)	(-1.09)
Alignment	-0.030**	-0.058***	-0.057***	-0.043***	-0.055**	-0.059***
	(-2.54)	(-3.12)	(-2.97)	(-3.04)	(-2.51)	(-2.65)
% in cash	-0.012	-0.008	-0.003	-0.006	-0.003	-0.003
	(-1.00)	(-0.44)	(-0.17)	(-0.37)	(-0.15)	(-0.12)
Tender offer	0.018**	0.029***	0.026***	0.018**	0.035***	0.034***
	(2.53)	(2.87)	(2.63)	(2.18)	(2.97)	(2.85)
Log asset (A)	0.015*	0.026**	0.032**	0.014	0.019	0.023
	(1.67)	(2.03)	(2.48)	(1.33)	(1.30)	(1.53)
Log asset (T)	0.005**	0.009***	0.009***	0.006***	0.011***	0.012***
	(2.25)	(2.86)	(2.80)	(2.65)	(3.32)	(3.66)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.079	0.084	0.111	0.055	0.052	0.081
N	1,044	757	650	1,044	757	650

Panel B: Bid premia

Sample:	Public targets					
Dep. var.:	Bid premia (t-1 day)			Bid premia (t-1 week)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal	-0.060	0.023	0.047	-0.045	-0.008	0.043
	(-0.83)	(0.22)	(0.46)	(-0.74)	(-0.08)	(0.43)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.077	0.014	0.059	0.086	0.016	0.068
N	906	658	566	906	658	566

Table 11: Anticipation? This table replicates Columns (3) and (5) in Table 2, Panel A and Panel B for acquirers of public targets. The dependent variable *GapSignal* proxies for the acquirer's revealed technological gap. The dependent variables are the target's CARs over two event windows: CAR(-1,1) and CAR(-3,3), as well as their absolute values. Control variables are omitted for brevity; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: CAR(-1,1)

Sample:	Acquirers of public targets					
Dep. var.:	|CAR(-1,1)|		CAR(-1,1)			
Subsample:	All		CAR(-1,1) < 0		CAR(-1,1) ≥ 0	
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal	0.024***	0.039***	0.011	0.006	-0.038***	-0.066***
	(2.97)	(3.12)	(0.96)	(0.34)	(-3.25)	(-3.11)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	Yes	No	Yes	No
AInd × Tind × Year FE	No	Yes	No	Yes	No	Yes
Adj. R^2	0.189	0.249	0.265	0.250	0.202	0.195
N	1,044	650	460	265	555	303

Panel B: CAR(-3,3)

Sample:	Acquirers of public targets					
Dep. var.:	|CAR(-3,3)|		CAR(-3,3)			
Subsample:	All		CAR(-3,3) < 0		CAR(-3,3) ≥ 0	
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal	0.025**	0.051***	0.005	0.011	-0.064***	-0.078***
	(2.53)	(3.37)	(0.34)	(0.35)	(-3.85)	(-2.69)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	Yes	No	Yes	No
AInd × Tind × Year FE	No	Yes	No	Yes	No	Yes
Adj. R^2	0.167	0.183	0.050	0.005	0.094	0.068
N	1,044	650	460	265	555	303

Table 12: Redefining the gap measure. This table replicates Columns (3) to (5) from both panels of Table 2, while redefining the dependent variable *GapSignal*—the proxy for the acquirer's revealed technological gap—based on CPC subclasses in Panel A, and 5-year patenting histories of the target and the acquirer's industry in Panel B, and weighting each patent by its application cohort-adjusted citation or the logarithm of its real economic value in Panels C and D, respectively. The dependent variables are the acquirer's CARs over two event windows: CAR(-1,1) and CAR(-3,3). Control variables are omitted for brevity; a complete list is provided in Section 3.2.3. $AInd$ and $TInd$ denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: *GapSignal* defined on finer patent categories

Sample:	Public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal (subclass)	-0.022*	-0.049***	-0.048***	-0.037***	-0.060***	-0.068***
	(-1.92)	(-2.86)	(-2.79)	(-2.76)	(-2.99)	(-3.40)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.078	0.081	0.111	0.053	0.057	0.091
N	1,044	757	650	1,044	757	650

Panel B: *GapSignal* defined on shorter patenting history

Sample:	Public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal (5-year)	-0.017	-0.044**	-0.042**	-0.030**	-0.053**	-0.057***
	(-1.41)	(-2.46)	(-2.33)	(-2.16)	(-2.52)	(-2.62)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.040	0.050	0.053	0.057	0.057	0.091
N	1,044	1,044	1,044	757	757	650

Table 12: Redefining the gap measure (continued).

Panel C: *GapSignal* computed with citation weighting

Sample:	Public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal (citation-weighted)	-0.018*	-0.038***	-0.035**	-0.032***	-0.045***	-0.045***
	(-1.92)	(-2.61)	(-2.34)	(-2.75)	(-2.76)	(-2.72)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.078	0.079	0.106	0.053	0.053	0.081
N	1,044	757	650	1,044	757	650

Panel D: *GapSignal* computed with value weighting

Sample:	Public targets					
Dep. var.:	CAR(-1, 1)			CAR(-3, 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal (value-weighted)	-0.013	-0.025*	-0.026**	-0.018*	-0.028*	-0.036**
	(-1.53)	(-1.91)	(-1.98)	(-1.67)	(-1.70)	(-2.31)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.077	0.072	0.103	0.049	0.046	0.079
N	1,044	757	650	1,044	757	650

Table 13: Target revaluation. This table replicates Table 10, while introducing additional regressors. *GapSignal* and *GapSignal* (T) measure the revealed technological gap of the acquirer and the target, respectively. *Alignment* and *Alignment* (T) measure the observable technological standing relative to the industry technology set of the acquirer and the target, respectively. The Dependent variables are the target's CARs over two event windows: CAR(-1,1) and CAR(-3,3). Other control variables are omitted for brevity; a complete list is provided in Section 3.2.3. *AInd* and *TInd* denote the 2-digit SIC industries of the acquirer and the target, respectively. All other variables are defined in Table A2. T-statistics are reported in brackets. Standard errors are robust to heteroskedasticity. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Sample:	Public targets					
Dep. var.:	CAR(-1,1)			CAR(-3,3)		
	(1)	(2)	(3)	(4)	(5)	(6)
GapSignal	-0.006	-0.048	0.282	-0.018	-0.100	0.110
	(-0.09)	(-0.27)	(0.79)	(-0.27)	(-0.56)	(0.30)
GapSignal (T)	-0.156**	-0.229	-0.150	-0.141*	-0.245	-0.327
	(-2.08)	(-1.33)	(-0.65)	(-1.73)	(-1.37)	(-1.33)
Alignment	0.080	0.179	0.092	0.015	0.176	0.252
	(1.14)	(1.03)	(0.41)	(0.18)	(0.97)	(1.09)
Alignment (T)	0.068	0.139	-0.195	0.064	0.160	-0.061
	(1.16)	(0.78)	(-0.55)	(1.00)	(0.94)	(-0.17)
Firm, deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, AInd, TInd FE	Yes	No	No	Yes	No	No
AInd × Year, Tind × Year FE	No	Yes	No	No	Yes	No
AInd × Tind × Year FE	No	No	Yes	No	No	Yes
Adj. R^2	0.079	0.084	0.111	0.055	0.052	0.081
N	1,044	757	650	1,044	757	650